Verified Stopping Dominates Adaptive Allocation in Multi-Container DNA Storage: A Leak-Free Computational Study on a Digital Twin and Real Read Streams

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Abstract

When an erasure-coded DNA archive is spread over several separately synthesised and handled containers of unknown quality, retrieval involves two decisions: when to stop sequencing, and which container to read next. We find that the first is worth an order of magnitude more than the second. All results are computational: no DNA was synthesised or sequenced for this paper, and the real-read experiments replay public Illumina runs. On the DT4DDS digital twin with six independently synthesised, aged and amplified containers, a streaming decoder that stops only when every stripe, its CRC-32 and the file's SHA-256 verify consumed 3.49k reads per retrieval (95% CI 3.37–3.60k). Without stopping, the same read proportions must *reserve* 4.80–5.15k reads to succeed on 99% of retrievals, so stopping reduces the bill by 27–32% and requires no model of the containers. The choice of container is a much smaller and less stable effect. Under the pre-registered configuration (a single efficiency multiplier per container and a beta-binomial likelihood with ρ_U=0.1), every model-based rule (Thompson sampling, UCB, learn-then-commit, information-greedy) read 7% to 441% *more* than round-robin when one container was near-dead. This penalty depends on the likelihood. At ρ_U=0.03, the value chosen by a corrected selection, four of the seven rules are statistically indistinguishable from round-robin (-1.2% to $+1.5\%$) and three remain clearly worse; no adaptive rule beat round-robin by more than a few per cent in any configuration we tested. *Fixed* read proportions with the same stopping rule were 4.2% cheaper than round-robin (pre-specified plan, 95% CI 3.5–4.9%), and proportions estimated from ten disjoint calibration replicates instead of hindsight performed as well, so the useful allocation knowledge can be obtained offline. A $2 \times 2 \times 2$ ablation shows that the penalty requires the frozen *combination* of container model, likelihood and prior width; relaxing any one of them places every rule 0 to 4.5% below round-robin, where fixed proportions already lie. Adding a dropout parameter, the physically meaningful change of the three, helps most where loss dominates. On the baseline it brings UCB to -3.5% (a separate, non-frozen configuration), which is comparable to the fixed plan rather than additional to it. At the 0.99-quantile, three adaptive rules with the dropout model were 1–9% below round-robin, but none of these reductions survives a cluster-bootstrap interval; fixed proportions reduced it by 450 reads (-8.7%) with an interval excluding zero. Over 62,310 end-to-end retrievals, covering every configuration in the paper except the clean-reproduction re-runs, no output file was wrong, and no wrong payload passed the inner checks in 77.3 million decode attempts. Because these retrievals share 580 synthesis replicates, we report both the per-retrieval upper limit (5.9×10^{-5}) and the per-synthesis limit (6.3×10^{-3}). A replay of real Illumina streams from two vendor pools, with calibration and evaluation reads kept disjoint, gives the same ranking. It also contains the one case in which allocation pays substantially: a prior calibrated on held-out reads of the same containers identified the weak container before retrieval and saved 16% per retrieval, but it required 1.8M calibration reads, 2.8 times the cost of one retrieval, and so breaks even only after about 17 retrievals of the same archive.

Index Terms

DNA data storage, adaptive sequencing, verified stopping, erasure coding, coverage depth, digital twin, Bayesian calibration, multi-armed bandits, reproducibility

I. Introduction

DNA has been demonstrated as an archival medium at kilobyte to hundreds-of-megabyte scale [1]–[5]. Retrieval requires sequencing, which samples molecules at random: some strands are read many times, others never, and

reads carry substitution, insertion and deletion errors [6], [7]. Systems therefore protect each strand with an inner code and groups of strands with an outer erasure code, and sequence to a fixed depth expected to recover enough strands [3], [5]. The number of reads needed for decoding—the *coverage depth*—is now studied in its own right [8], [9].

We consider archives split across several physically separate *containers* (vials, capsules, pools) that are synthesised, stored and handled separately. This is a working assumption rather than established practice; random access so far addresses files within one pool [5], [10], and recent theory treats erasure-coded data partitioned over containers with uniform sampling [11] and container failures [12]. Containers age, are amplified and fail differently, and these differences are unknown when a retrieval starts. Two operational questions follow: *when* to stop sequencing, and *which* container to sequence next.

Problem. Given an erasure-coded file distributed over M containers of unknown, heterogeneous quality, some of which may have failed, sequence in batches and decide after each batch which container to read and whether to stop, so that the output file is byte-identical to the stored one with high probability while the number of reads is small.

Findings. (i) *Verified stopping provides the robust saving, measured as reads consumed against capacity reserved.* It needs no container model and never returned a wrong file. On the digital twin it turns the 4.80–5.15k reads that a static plan must reserve into a mean of 3.49k (-27 to -32%); on the two real-read pools it reduced reads relative to fixed depth by 14% and 25%. (ii) *Allocation is worth a few per cent at best and is harmful under a misspecified container model.* Under the frozen configuration, every model-based rule read 7–441% more than round-robin when a container was near-dead. The penalty depends on the likelihood: at the ρ_U chosen by a corrected selection, four of the seven rules are statistically indistinguishable from round-robin and three remain clearly worse (Section VI-J). An ablation attributes the penalty to the controller being slow to recognise a bad container; each of three independent modelling choices is sufficient to cause it, and relaxing any one of them removes it. Every rule then lies 0–4.5% below round-robin, where fixed proportions already lie (-4.2% for the pre-specified plan, and the same within noise when the proportions are estimated from disjoint calibration replicates rather than in hindsight). The only substantially larger allocation effect requires identifying the weak container *before* retrieval from calibration reads (-16% per retrieval on one real-read pool, repaid after about 17 retrievals). (iii) *No adaptive rule reliably lowered the 0.99-quantile.* Three configurations with the dropout model are 1–9% below round-robin, but every one of those intervals contains zero, whereas fixed proportions lower it by 8.7% with an interval that excludes zero. Online learning therefore gave no demonstrated reduction in reserved capacity. (iv) *Two zero-success results reflect limits of the archive design rather than of the policies.* Two failed containers leave exactly K fragments per stripe by construction, and under the photolithographic synthesis model no policy or static plan in our design recovered the file.

Scope. All work is computational; nothing was synthesised or sequenced for this paper. We use the term *exact retrieval* only for our digital-twin pipeline, where the output file is checked byte for byte against the stored file. On replayed real reads, decodability is an oracle event and we speak of *recovery*.

Contributions.

1) *A measurement of the value of verified stopping.* A verification-gated streaming pipeline (Section IV) is evaluated against a static family that includes a hindsight-tuned plan and a calibration-estimated plan, with reads consumed reported separately from capacity reserved (Section VI-B). Stopping when decodable [13] and checksum-verified decoding [4] are both known; our contribution is to quantify their value for heterogeneous multi-container retrieval, end to end to verified bytes.
2) *A negative result for adaptive allocation, with its cause isolated and repaired.* Eight rules, including Thompson sampling, UCB, learn-then-commit and information-greedy design [14]–[17], lose to round-robin under a near-dead container. A $2 \times 2 \times 2$ ablation separates the container model from the likelihood and the prior (Section VI-E), and a two-parameter model with dropout removes most of the penalty (Section VI-F). The failure therefore lies in the model rather than in adaptivity, and the experiments bound what a better model gains.
3) *A reproducible end-to-end evaluation with explicit uncertainty.* The full chain from DT4DDS reads [7] of independently synthesised containers to verified bytes is run under pre-registered protocols. We report safety

bounds per retrieval and per synthesis replicate, tail metrics with cluster-bootstrap intervals, sensitivity to the prior, the consensus cap, six further archive geometries and five failure modes, and a cost model in which reads are not the only charge. Every table and figure is regenerated from the released artifact (Section IX).

II. Background and Related Work

DNA storage systems and channels. Storage systems combine constrained encodings with inner and outer codes [2]–[5], [18], [19]. Coverage is over-dispersed, a few percent of strands drop out, and errors arise mostly in synthesis and sequencing [6], [7], [20]. DT4DDS [7] is a digital twin of the synthesis, aging, PCR and sequencing steps whose error and bias parameters were fitted to experiments; we use it as the simulator of our primary evidence. Information-theoretic models treat the channel as sampling with replacement followed by noise [21]–[23].

Coverage depth. For a sampler that cannot be controlled, MDS codes and uniform sampling minimise the expected number of reads [8]. Follow-ups study random access [9], real-data sampling distributions [24], single-container coverage depth of structured codes [25], and the reliability of MDS-coded retrieval [26]. Cao *et al.* [27] drop the uniform-sampling idealisation that most of this line assumes: they model the PCR and sequencing channel as nonuniform, instantiate it log-normally, and analyse the MDS coverage-depth problem for both noiseless and noisy channels, deriving the expected minimum depth for complete decoding in the noiseless case and, for the noisy case, a coupon-collector envelope and a critical-quantile refinement, with sequencing sample-size estimates validated by Monte-Carlo simulation. Their object is one pool: how deep to sequence it so that enough *strands* are recovered. That is the quantity our controller consumes rather than derives — our consensus-success curve and over-dispersed coverage model play the role their channel model plays — and it is complementary to what we study: several independently handled containers of unknown and heterogeneous quality, some of which may have failed; *which* container to read next after each batch; when to stop, decided by verification rather than by a depth computed in advance; and, in the digital-twin experiments, byte-for-byte verification of the returned file. We take nonuniform sampling, coverage-depth analysis and the use of checksums as established by this literature, not as contributions of ours.

Distributed coverage depth. Two 2026 papers are closest to our setting. Kong *et al.* [11] define the distributed coverage-depth problem: the n columns of a rank-k generator matrix are split over M containers by an ordered partition, and in every round *one* column is drawn uniformly at random, with replacement and *noiselessly*, from *each* container; $\tau(G, \mathcal{P})$ is the first round at which the drawn columns span $\mathbb{F}_q^k$. They derive an exact formula for $\mathbb{E}[\tau(G, \mathcal{P})]$ for arbitrary linear codes and partitions by q-binomial Möbius inversion, prove that an $[n, k]_q$ MDS code is optimal for every fixed partition (in rounds and in total reads), give the lower bound $M\,\mathbb{E}[\tau] \geq M\lceil k/M \rceil$ with $\min_{M,\mathcal{P}} M\,\mathbb{E}[\tau] = k$ attained by a staircase k-partition, characterise one-round-perfect layouts, separate the regimes in which distribution genuinely lowers total reads (constant-size containers) from those in which it only buys parallelism (containers of size k^{ε}), and prove that the q-ary simplex code uniquely minimises single-container coverage depth, resolving a conjecture of Bertuzzo *et al.* [25]. Their model is deliberately idealised: sampling is uniform within and across containers, containers are homogeneous and never fail, reads are noiseless, every round costs M reads whether or not they are needed (their Remark II.1), and the allocation is fixed in advance. Their Section V lists "nonuniform, asynchronous, adaptive, and noisy sampling" as open problem 4 and an asynchronous variant with a total read budget as an open problem. Our work is an empirical study of that gap: heterogeneous, noisy, possibly failed containers, reads allocated one batch at a time, and stopping decided by verification rather than by an oracle span condition. It is complementary to theirs; we prove nothing about optimal codes or layouts, and we use their regime distinction (parallelism versus genuine read saving) to interpret our results. Levy *et al.* [12] study a different objective: reconstructing the contents of one *failed* container from the surviving ones, with scalar MDS and MDS array codes. Full-message retrieval, which we study, is still required after such a repair, and the two stopping rules differ (a prescribed missing block versus the whole message). Neither work allocates reads adaptively, uses noisy reads, or decodes to bytes. These works analyse *fixed* sampling laws; we study a *controllable* sampler empirically.

Adaptive sequencing and sequential decisions. Nanopore adaptive sampling selects molecules in real time [28]–[30], and DNA-storage readouts have been sequenced until sufficient coverage [13]. Choosing a container resembles a multi-armed bandit with a combinatorial (K-of-N per stripe) reward; we use Thompson sampling [14], [31], upper

TABLE I

POSITIONING AGAINST THE CLOSEST PRIOR WORK. Y: YES; N: NO; P: PARTIAL; I: IMPLICIT (EXPECTED READS UNTIL DECODABILITY ANALYSED, NO OPERATIONAL STOPPING RULE); –: NOT STATED. "CONTAINERS" ARE STORAGE NODES IN [37] AND SAMPLING SOURCES IN [45].

Work	Multi-container	Container failures	Heterog. containers	Correlated errors	Adaptive read allocation	Stopping rule	Real reads	End-to-end digital twin
Lopez et al. 2019 [13]	N	N	N	–	N	P	Y	N
Gimpel et al. 2023 [7]	N	N	P	N	N	N	Y	Y
Bar-Lev et al. 2025 [8]	N	N	N	N	N	I	N	N
Gruica et al. 2024 [9]	N	N	N	N	N	I	N	N
Cao & Chen 2025 [24]	N	N	N	N	N	I	P	N
Cao et al. 2026 [27]	N	N	N	Y	N	I	P	N
Kas Hanna 2026 [26]	N	N	N	N	N	N	N	N
Levy et al. 2026 [12]	Y	Y	N	N	N	I	N	N
Kong et al. 2026 [11]	Y	N	N	N	N	I	N	N
Leong et al. 2012 [37]	Y	Y	N	N	N	N	N	N
Weilguny et al. 2023 [30]	N	N	–	N	Y	–	Y	N
Bubeck et al. 2013 [45]	Y	N	Y	N	Y	N	N	N
This work	Y	P	Y	P	Y	Y	P	Y

Notes on "this work": container failures are simulated (digital twin) or injected (replay), not observed; correlated errors are validated on real nanopore reads but enter the controller only through consensus curves; real reads (ENA PRJEB65931 [46], same-batch aliquots) are used in a semi-empirical replay with oracle decodability, so they are marked partial; the end-to-end digital twin is DT4DDS [7]. Gimpel et al. [7]: heterogeneity characterised across handling conditions of one pool, and the digital twin itself. Cao & Chen [24] and Cao et al. [27]: log-normal nonuniform channel fitted to sequencing data; [27] treats noisy as well as noiseless MDS coverage depth, so its "correlated errors" cell is Y in the sense of a noisy channel model, but it is a single-pool, strand-level analysis with no containers, failures, allocation or operational stopping rule. Weilguny et al. [30]: genomic loci, not storage containers. Kong et al. [11] (full paper, 33 pp.): containers are homogeneous, never fail and are sampled noiselessly and uniformly, one read per container per round, with the allocation fixed in advance; adaptive, nonuniform, asynchronous and noisy sampling are listed there as open problems. Levy et al. [12]: objective is repairing one failed container from the survivors, not full-message retrieval. Neither derives an operational stopping rule, so both are marked I.

confidence bounds [15], explore-then-commit [16] and an information-greedy rule from Bayesian experimental design [17] as baselines (see [32]).

Placement and consensus. Erasure-coded storage limits the fragments of a stripe per failure domain [33]–[37]. Strands are reconstructed from noisy reads by trace reconstruction [38]–[40] after clustering [41] or learned models [42]; systematic nanopore errors are documented [43], [44]. Table I positions this work.

Relation to prior work. Prior DNA-storage work already stops sequencing when the decoder has enough information [13] and verifies decoded data with checksums [4], and the coverage-depth literature analyses the expected number of reads until decodability, including across containers [8], [11], [12]. Stopping, verification and distributed DNA read allocation are therefore not contributions of this paper. We did not find in that literature a measurement, on an end-to-end pipeline that decodes to verified bytes, of how much a verification-gated stop is worth relative to the capacity a static plan must reserve when containers are heterogeneous and may have failed, or of whether choosing *which* container to read next with a learned container model helps or hurts under those conditions. We provide that measurement, the mechanism behind the negative allocation result, and an artifact from which every number can be regenerated.

III. SYSTEM MODEL

Table II lists the notation.

A. Storage

A file is bundled, hashed (SHA-256) and split into S stripes of K data fragments of P bytes. A systematic Cauchy Reed–Solomon code [47] over GF(2^8) extends each stripe to N fragments, any K of which suffice. Each fragment becomes a strand carrying an address, an address check byte, the payload and an inner Reed–Solomon code with four parity bytes plus a CRC-16, mapped to DNA by a whitened 2-bit code. Code parameters, a CRC-32 per stripe and the SHA-256 are kept in a manifest outside the DNA. Fragments are assigned round-robin to M containers.

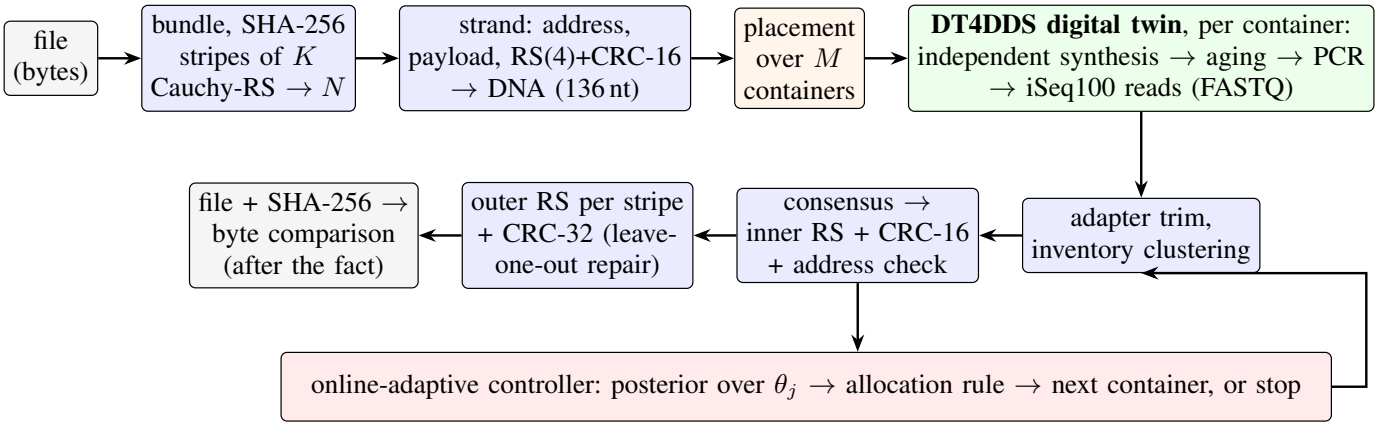


Fig. 1. Pipeline of the primary experiments (conceptual diagram). Every container is an independent DT4DDS synthesis with its own handling history; its reads are consumed in batches by the same streaming decoder that reads FASTQ files. The true file is used only after a retrieval ends, to count wrong outputs and false accepts.

TABLE II
NOTATION (CODE OBJECTS IN PACKAGE D3RA).

Symbol	Meaning	Code
K, N, S, P	data / total fragments per stripe, stripes, payload bytes	`erasure_codes`
M, j	number of containers, container index	`Layout`
n_j	fragments stored in container j	`Layout`
b	batch size (reads per decision)	`AdaptiveConfig`
R_j, U_j	reads taken from j, verified fragments from j	`StreamRetriever`
θ_j, Θ, s	efficiency multiplier, its grid, prior log-sd	`AdaptiveConfig`
f_j	prior probability that container j failed	`ChannelParams`
$\bar{\pi}_j$	prior per-fragment read share of container j	`Adaptive Controller`
k, d, $\varphi(t)$	NB shape, dropout, consensus-success curve	`ChannelParams`
$Q(\mu)$	P(fragment decodable) at mean coverage μ	`_QTable`
ρ_U	beta-binomial ICC of recoveries in the likelihood	`overdispersion`
h_{ij}	hazard of recovery within the next batch	`scores`
q	posterior quantile for UCB (q) and tail-aware ($1-q$)	`quantile`
ν_s	remaining need of stripe s (K minus verified)	`scores`
w_j, e_j, α	static proportions, hindsight per-read recovery rate, exponent	`static_family`
θ_j^*	calibration target (best-fitting multiplier)	`_theta_star`
q99	0.99-quantile of reads per retrieval	`dt_analysis`

B. Sequencing and recovery model used by the controller

Reads from container j form a stream. The controller's model (used only to rank containers and to forecast; never to decide that a fragment is recovered) treats the reads of a stored fragment as negative-binomial with mean $\theta_j R_j \bar{\pi}_j$ after R_j reads, where $\bar{\pi}_j$ is the prior per-fragment read share (yield over stored fragments) and θ_j an unknown efficiency multiplier (θ_j=0: failed container). A fragment becomes decodable after t reads with probability $\varphi(t)$ (consensus-success curve), and a molecule is absent with probability d (dropout). Integrating over abundance gives

the probability $Q(\mu)$ that a stored fragment is decodable at mean coverage μ:

$$Q(\mu) = (1-d)\,\mathbb{E}_g\Big[\sum_{n\geq 0} \mathrm{NB}(n; k, g\mu)\,\varphi(n)\Big], \tag{1}$$

with a shared log-normal strand effect g (Gauss–Hermite quadrature). Every container starts from the same generic literature prior (yield 0.9, NB shape k=10, dropout 0.5%, a generic consensus curve, failure probability 10^{-3}); no calibration data of the evaluated system is used in the primary experiments.

C. Objective and metrics

A policy chooses, after each batch of b reads, a container and whether to stop. We report the following quantities separately:

1) **Mean reads over all attempts** $\bar{r}_{\text{all}}$: the average reads consumed per retrieval *attempt*, counting attempts that end in failure (which consume the whole stream). This is what a pay-per-read user pays on average.
2) **Mean reads conditional on success** $\bar{r}_{\text{ok}}$: the average over attempts that returned the verified file. It is the right metric only when the success probability is the same for the policies compared, and it says nothing about the cost of failures.
3) **Success probability** p_{ok}: the fraction of attempts that returned the verified file within the stream.
4) **Tail quantiles** q95 and q99 of the consumed reads, with cluster-bootstrap intervals.
5) **Reserved capacity** $B_{0.99}$: the budget a plan must commit *in advance* to succeed on 99% of retrievals, i.e. the q99 of its read requirement. Where every retrieval succeeds, $B_{0.99}$ coincides numerically with (4); the two differ in who pays it. A user who reserves a flowcell pays $B_{0.99}$ on every retrieval, whether or not the retrieval needed it, whereas a pay-per-read user pays (1).

Verified stopping replaces the reserved capacity $B_{0.99}$ with the consumed mean $\bar{r}_{\text{all}}$; its saving is therefore a comparison between a policy that stops and a plan that does not. Comparing two policies that both stop is a separate, allocation-only question (Section VI-B). No metric here is a monetary cost; a cost model with per-run and per-container charges is given in Section VI-M.

IV. Online-Adaptive Retrieval

A. Posterior over container efficiency

The support of θ_j is $\Theta = \{0\} \cup \{\theta^{(1)}, \ldots, \theta^{(25)}\}$, log-spaced in $[0.01, 5]$, with prior $p_0(0) = f_j$ and $p_0(\theta^{(\ell)}) \propto (1-f_j)\exp(-\frac{1}{2}(\ln\theta^{(\ell)}/s)^2)$, with $s = 0.5$ in every policy run. After R_j reads with U_j verified fragments among the n_j stored in container j, we use a beta-binomial likelihood with intra-container correlation ρ_U:

$$p(\theta \mid R_j, U_j) \propto p_0(\theta)\,\mathrm{BB}\big(U_j;\, n_j,\, Q(\theta R_j \bar{\pi}_j),\, \rho_U\big), \tag{2}$$

where $\mathrm{BB}(u; n, p, \rho)$ has mean np and variance $np(1-p)(1+(n-1)\rho)$; ρ_U=0 recovers the binomial. Recoveries of a container's fragments are correlated (shared synthesis bias, dropout), so the binomial is over-confident; ρ_U is chosen on disjoint calibration replicates (Section V-D). We call the controller *online-adaptive*, not self-calibrating: it adapts to observations, but its uncertainty is not well calibrated (Section VI-I).

B. Allocation rules

For an unrecovered fragment i in container j, the hazard of becoming decodable within the next b reads is $h_{ij} = [Q(\tilde{\theta}_j(R_j{+}b)\bar{\pi}_j) - Q(\tilde{\theta}_j R_j \bar{\pi}_j)]/[1 - Q(\tilde{\theta}_j R_j \bar{\pi}_j)]$, where $\tilde{\theta}_j$ is chosen by the rule:

- *Round-robin + verified stop*: reads every container in turn (no model).
- *Coverage-greedy*: $\tilde{\theta}_j$ = posterior mean; reads $\arg\max_j \sum_i h_{ij}$ over fragments of unfinished stripes.
- *Pivotal*: weights each fragment by a normal-approximation sensitivity of its stripe's completion probability (our earlier rule; details in the supplement).
- *Thompson sampling* [14], [31]: $\tilde{\theta}_j$ drawn from the posterior; coverage score.
- *UCB* [15]: $\tilde{\theta}_j$ = posterior 0.9-quantile.
- *Learn-then-commit* [16]: five round-robin batches per container, then the posterior is frozen and the coverage score used.

- *Info-greedy* (Bayesian experimental design [17]): reads the container with the largest posterior variance of $\ln \theta_j$ until all fall below 10^{-3}, then coverage-greedy.
- *Tail-aware*: $\tilde{\theta}_j$ = posterior 0.1-quantile (pessimistic) and the score counts only the 10% of stripes with the largest remaining need (the bottleneck stripes that set the retrieval's length); a heuristic aimed at the q99.

All rules probe each container with one batch first. Table III gives the operational specification of the loop and of every rule — initialisation, batch size, tie-breaking, when the posterior is updated, the stopping test and how an exhausted stream is handled. The *static* family reads fixed proportions $w_j \propto e_j^{\alpha}$, $\alpha \in \{0, 0.5, 1, 2\}$, until decodable, where e_j is the per-read recovery rate of container j measured in hindsight on the same scenario: α=0 is uniform fixed depth, α=1 equalises expected coverage, and the best α is a hindsight-tuned ("oracle") static plan.

C. Physical interpretation of container switching

Our controller picks a container after every batch of b=50 reads. This does not assume that a sequencer can be redirected between physical vials that often, and the experiments do not demonstrate a physical sequencing controller. They model an *allocation policy over M independent read streams*: each container has its own prepared library and its own stream of reads, and the policy decides from which stream the next batch is consumed. Three physical arrangements realise this abstraction at different costs. (i) *Multiplexed run*: all M libraries are barcoded and loaded on one flow cell; reads arrive interleaved and "choosing a container" means choosing which barcode's reads to keep processing. This matches our model closely but does not reduce the reads the instrument actually produces, so it saves analysis and, on pay-per-read services, money, but not instrument time. (ii) *Streaming (nanopore)*: reads arrive in real time, and adaptive sampling can reject molecules mid-read [28], [29]; with one flow cell per container, stopping is directly actionable, and with barcoded libraries selective sequencing approximates container choice. (iii) *Sequential runs*: one run per container, where a switch costs a library preparation and a run setup. Our frozen cost metric (reads) describes case (i) with per-read billing and is optimistic for cases (ii) and (iii); the per-batch and per-container charges in Section VI-M bound how much that optimism matters. Verified stopping, which is our main recommendation, needs no switching mechanism; the decoder only has to signal that the file is complete.

D. Verified stopping

A retrieval stops with an output file only when (i) every stripe holds K fragments that each passed inner RS decoding, CRC-16 and the address check, (ii) every stripe matches its CRC-32, and (iii) the file matches its SHA-256. If (ii) fails, one more fragment is requested for that stripe and the faulty fragment is localised by leave-one-out decoding. If the streams are exhausted first, the retrieval reports failure; it never returns an unverified file. An early-abandonment (futility) rule is disabled in all runs.

E. Completion forecast

The controller also forecasts the probability of finishing within a remaining budget. The *analytic* forecast multiplies per-stripe normal approximations of $\mathbb{P}(\sum_{i \in s} X_i \geq \nu_s)$ with the remaining budget spread evenly over containers. The *posterior-simulation* forecast (default) draws 200 values of θ from the posterior, spreads the remaining budget in proportion to the reads spent so far, simulates per-fragment recoveries and reports the fraction of draws in which every stripe completes.

V. Evaluation Methodology

A. Digital-twin end-to-end pipeline (primary evidence)

We store a fixed 18,432-byte random file with P=24-byte payloads, $(K, N) = (32, 48)$, over M=6 containers. The stripe count follows from what is actually encoded: the library bundles the file with a 162-byte header (name, length, hash) into an 18,594-byte blob, and $\lceil 18{,}594/(32 \times 24) \rceil = 25$ stripes of 32 data fragments cover it, the last one zero-padded (24 stripes would hold only 18,432 bytes, which is the payload alone). This gives $25 \times 48 = 1{,}200$ strands of 136 nt, 200 per container, 8 fragments of every stripe per container, and, at 60 simulated reads per stored strand, a maximum stream of $60 \times 1{,}200 = 72{,}000$ reads. All counts reported are those produced by the code rather than rounded design targets. For each container we run DT4DDS 1.1.0 [7] with its published default parameters: array synthesis (Twist model; log-normal oligo counts, σ=0.3), sampling of a given number of copies per strand,

TABLE III

OPERATIONAL SPECIFICATION OF THE RETRIEVAL LOOP AND OF EVERY ALLOCATION RULE. ALL RULES SHARE THE LOOP, THE BATCH SIZE, THE PROBE, THE TIE-BREAKING AND THE STOPPING TEST; THEY DIFFER ONLY IN THE SCORE σ_j AND IN WHICH POINT VALUE $\tilde{\theta}_j$ OF THE POSTERIOR THEY USE. R_j: READS SPENT ON CONTAINER j; U_j: FRAGMENTS OF CONTAINER j VERIFIED FROM ITS OWN READS; n_j: FRAGMENTS STORED IN j; h_{ij}: HAZARD THAT UNRECOVERED FRAGMENT i IN j BECOMES DECODABLE WITHIN THE NEXT BATCH; $\mathcal{O}$: FRAGMENTS OF STRIPES THAT STILL NEED FRAGMENTS.

Retrieval loop (identical for every policy)

1) $R \leftarrow 0$; pools empty; tracker empty.
2) *Probe*: read b=50 reads from every container that stores fragments, in index order (the static plans skip the probe and start from their proportions).
3) **repeat**
   a) *Stopping test*: if every stripe holds K fragments that passed inner RS, CRC-16 and the address check, outer-decode all stripes; if every stripe matches its CRC-32 and the blob matches its SHA-256, **return** the file. If a stripe fails its CRC-32, raise that stripe's requirement by one fragment (leave-one-out localisation) and continue; after 20 such rounds, stop and report failure.
   b) If every stream is exhausted, **return** failure (never an unverified file).
   c) Compute the posterior (2) from (R_j, U_j) for every container, take $\tilde{\theta}_j$ as the rule prescribes, score the available containers and read b reads from $\arg\max_j \sigma_j$; ties break to the lowest index.
   d) Cluster the new reads against expected address prefixes, form a consensus from at most 24 reads per cluster, decode, and update U_j and the tracker. Reads are charged even if the stream runs dry.

Allocation rules: score σ_j

- *Round-robin + verified stop*: σ_j unused; container (round $\bmod M$). No model is evaluated.
- *Static* α: no model; read the container minimising R_j/w_j with $w_j \propto e_j^{\alpha}$ (exhausted containers excluded; if all positive-weight containers are exhausted, read the rest so the failure is charged).
- *Coverage-greedy*: $\tilde{\theta}_j = \mathbb{E}[\theta_j]$; $\sigma_j = \sum_{i \in \mathcal{O} \cap j} h_{ij}$.
- *Pivotal*: $\tilde{\theta}_j = \mathbb{E}[\theta_j]$; $\sigma_j = \sum_i w_s(i)\, h_{ij}(1 - \rho_i + h_{ij})$ with w_s a normal-approximation sensitivity of stripe s's completion probability.
- *Thompson sampling*: $\tilde{\theta}_j \sim p(\theta_j \mid R_j, U_j)$, redrawn every round; coverage score.
- *UCB*: $\tilde{\theta}_j$ = posterior 0.9-quantile; coverage score.
- *Learn-then-commit*: round-robin for 5 batches per container, then freeze the posterior and use the coverage score with $\tilde{\theta}_j = \mathbb{E}[\theta_j]$ from the frozen posterior.
- *Info-greedy*: $\sigma_j = \mathrm{Var}[\ln \theta_j]$ while any container has $\mathrm{Var}[\ln \theta_j] > 10^{-3}$, then coverage-greedy. Containers with no unrecovered fragment score 0.
- *Tail-aware*: $\tilde{\theta}_j$ = posterior 0.1-quantile; score counts only fragments of the 10% of unfinished stripes with the smallest standardised slack, falling back to the coverage score when no such fragment can be advanced.

Container model. Frozen: $\theta_j \in \{0\} \cup \{0.01 \ldots 5\}$ (25 log-spaced points), prior $p_0(0) = 10^{-3}$ and log-normal with s=0.5 otherwise, beta-binomial likelihood with ρ_U=0.1. Revision 4 adds the two-parameter model: $d_j \in \{0, 0.05, 0.15, 0.3, 0.5, 0.7, 0.9\}$ with $p_0(d_j{=}0) = 0.6$ and the rest uniform, a joint posterior over (θ_j, d_j), verified-fragment probability $(1 - d_j)Q(\theta_j R_j \bar{\pi}_j)$, and hazards multiplied by $(1 - \hat{d}_j)$ so that a fragment still missing after many reads is increasingly attributed to permanent dropout.

aging, high-fidelity PCR and Illumina iSeq100 single-end 150 nt sequencing, producing 60 reads per stored strand. The six containers are *independent* syntheses with their own seeds and histories. The reads, after adapter trimming, go through inventory clustering against the expected address prefixes, consensus (at most 24 reads per cluster), inner decoding, outer decoding with CRC-32 repair and SHA-256 verification, in batches of 50 reads. Every container decodes its own clusters; nothing uses the stored strands to decide recovery. The stored file and payloads are used only after the retrieval to count wrong outputs and false accepts (a wrong payload that passed the inner checks).

B. Scenarios

Baseline: fresh; aged one half-life; aged two half-lives with 10 copies per strand; 30 PCR cycles; 45 PCR cycles with 20 copies; near-dead (two half-lives, 40 cycles, σ=0.5, 3 copies). *One/two failed*: the fresh (and the 30-cycle) container returns only non-informative reads. *Homogeneous control*: six fresh containers. *Photolithographic*: the baseline histories with DT4DDS's photolithographic synthesis model. *Severity sweep*: the baseline with the near-dead container's copy number set to 200, 30, 10, 3 or 1. *Additional failure modes*: half of one container's designs physically lost before storage; a quarter lost in each of two containers; one container whose reads are 90% not derived from stored strands (library-preparation failure or severe contamination); another at 50%; and a common-cause scenario in which every container is aged two further half-lives and amplified harder. *Additional geometries*: 3 and 12 containers, $(32, 40)$, $(32, 64)$ and $(16, 24)$ codes, and a file four times larger. With three containers the six conditions cannot all be kept; we retain one healthy, one weak and one near-dead container, so a third of that archive is near-dead instead of a sixth, which makes it a harder archive and not only a smaller one.

C. Frozen protocol and statistics

Design, policies, metrics, statistics, pre-specified comparisons and the calibration rule were written to `dt_protocol.yaml` before the final runs; deviations are listed in Appendix B. Each scenario uses 30 independent syntheses of all six containers $\times$ 10 read orders (300 retrievals per policy); seeds are derived with CRC-32 of the scenario, replicate and container; retrieval is bit-for-bit reproducible given the committed read streams, but DT4DDS itself is not seed-deterministic, so the streams are part of the artifact (Section IX). CIs are cluster bootstrap percentile intervals over syntheses (2,000 resamples [48]); comparisons are paired over the same syntheses and read orders; wrong-output and false-accept rates carry exact Clopper–Pearson bounds [49]. All pre-specified comparisons are reported, including negative ones, without significance stars.

D. Calibration protocol

ρ_U is chosen from $\{0, 0.003, 0.01, 0.03, 0.1\}$ on ten *disjoint* baseline replicates (seeds 1000–1009) as the value whose 90% credible interval covers the target θ_j^* closest to 90% of the time. θ_j^* is the multiplier that best fits the container's full recovery curve under the generic model, and the selection used prior sd $s{=}1$ whereas the policy runs use $s{=}0.5$ (a mismatch identified after freezing; see Section VI-I); intervals are evaluated at the read count where the prior predicts 40% recovery and at twice that. The selected value is then fixed as the default for every policy run, evaluated on the held-out replicates 0–29, and used on the real-read replay. The completion forecast is evaluated with a budget equal to the median coverage-greedy cost, at 30/50/70/85% of it, by expected calibration error (ECE, 10 bins), Brier score against a constant (climatology) forecast [50] and AUC.

E. Real read streams (secondary evidence)

We replay public Illumina iSeq runs of two pools from ENA PRJEB65931 [7], [46]: a Twist pool after 0/2/7 days of aging or 15/60/90 PCR cycles, and a Genscript pool (aging 0 d A/B, 2 d, 7 d; PCR 15/30 cycles). Each run emulates one container holding 250 stripes of $(32, 48)$, i.e. $250 \times 48 = 12{,}000$ designed strands. The Twist design file contains exactly 12,000 strands (108 nt), all of which are used. The Genscript design file contains 12,472 strands (102 nt); we draw a seeded random subset of 12,000 and *exclude* the remaining 472 designs entirely, discarding the reads that map to them: of the $6 \times 900{,}000$ evaluation reads, 86.3% map to a design and 4.23% of those mapped reads are discarded as excluded designs, leaving 4,461,804 usable reads. The excluded strands serve no other purpose, and the subset is the same in every condition and policy, so all Genscript comparisons use identical read material. We use 200 read permutations and a paired bootstrap.

Real and emulated components of the replay. The replay is *semi-empirical*. *Taken from the sequencing data*: the reads themselves, their number, which designed strand each read came from, the per-container yield and abundance profile that follows from those counts, the fraction of reads that map to no design, and the order in which reads arrive (we permute the real evaluation reads rather than generating arrivals from a model). *Supplied by us*: the partitioning of one pool's runs into "containers"; the erasure-coded layout of 250 stripes over those containers, which was never physically synthesised; and *the decision that a fragment has been recovered*. This decision is an *oracle* event: a fragment counts as recovered when the consensus of its first t reads in a random permutation of the run equals the designed strand, with t drawn from per-strand consensus curves estimated from real reads. Our inner Reed–Solomon code, the CRC-16, the address check, the outer code, the CRC-32 and the SHA-256 *are not executed* on this path, and no bytes are compared. The replay therefore supports no statement about wrong outputs, false accepts or byte-exact retrieval; those results come only from the digital twin, where the whole decode chain runs. Nor is the replay a physical demonstration of the system, since the containers are aliquot runs of one synthesis batch. It supports the comparison of *allocation and stopping policies* over read streams with real arrival statistics, and we use it only as secondary evidence on the ranking of policies.

Separation of calibration and evaluation reads. Each run is read once and split by position: the first 300,000 reads are the calibration split and the next 900,000 the evaluation split, so no read is in both. Everything that generates a recovery event in the evaluation comes from the evaluation split alone: the read order is a permutation of the evaluation reads, and the number of reads a strand needs is drawn against per-strand consensus curves. Two beliefs are used, and they differ exactly in which split they may see. The *generic* prior (all primary replay results) sees no data from these pools at all: yield 0.9, NB shape 10, dropout 0.5%, a generic consensus curve.

TABLE IV

SAFETY OF VERIFICATION-GATED STOPPING OVER EVERY DIGITAL-TWIN RETRIEVAL OF THIS PAPER, WITH THE DEPENDENCE BETWEEN RETRIEVALS MADE EXPLICIT. RETRIEVALS THAT SHARE A SYNTHESIS REPLICATE ARE NOT INDEPENDENT, SO WE ALSO REPORT THE BOUND THAT TREATS ONE SYNTHESIS AS ONE UNIT. UPPER LIMITS OF EXACT TWO-SIDED 95% CLOPPER–PEARSON INTERVALS.

Quantity	Value
Retrievals (all scenarios, policies, geometries)	62,310
wrong output files	0
Independent synthesis replicates	580
retrievals per replicate (mean)	107
synthesis replicates with a wrong output	0
Wrong-output rate per retrieval, 95% upper limit	5.9×10^{-5}
per synthesis replicate (conservative)	6.3×10^{-3}
Fragment decode attempts	77,290,829
false accepts (wrong payload passed inner checks)	0
attempts per synthesis replicate (mean)	133,260
False-accept rate per attempt, 95% upper limit	4.8×10^{-8}
per synthesis replicate (conservative)	6.3×10^{-3}

The *calibrated* belief (labelled "leak-free calibrated" in the tables) sees only the calibration split, i.e. per-container read counts and pooled consensus statistics estimated from calibration reads. Evaluation data is used deliberately in one place only: the hindsight-tuned static plan, which is labelled as an oracle baseline wherever it appears and serves as an upper bound on what a static allocation could achieve. Our first analysis of these pools did not keep the splits separate (per-strand consensus curves were estimated on the evaluation reads and static weights tuned on them); it is superseded, kept only as an upper bound in Appendix A, and no claim in this paper rests on it.

VI. RESULTS

A. Safety of verified stopping

Table IV reports the safety record together with the dependence structure behind it. It pools every digital-twin retrieval in this paper except the clean-reproduction re-runs (Section IX), which repeat replicates rather than adding independent ones. Across 62,310 end-to-end retrievals no output file differed from the stored file in a single byte, and in 77.3 million fragment decode attempts no wrong payload passed inner RS decoding, the CRC-16 and the address check. Unsuccessful retrievals (one or two failed containers) ended by exhausting the streams and reported failure instead of returning a file.

These retrievals are *not* independent experiments. They come from 580 distinct synthesis replicates, 107 retrievals per replicate on average (several read orders, policies, container models and prior settings applied to the same simulated DNA), and the decode attempts inside one retrieval share the same reads. A Clopper–Pearson bound that treats every retrieval as an independent Bernoulli trial would therefore overstate the evidence. Per retrieval the upper limit is 5.9×10^{-5}; treating one synthesis replicate as one unit, which is conservative because it discards the information in repeated read orders, it is 6.3×10^{-3}, two orders of magnitude weaker. The false-accept limits are 4.8×10^{-8} per attempt and 6.3×10^{-3} per synthesis replicate.

Zero observed errors does not imply zero error *probability*: the data are consistent with any rate below these bounds, which are of order 10^{-4} per retrieval and 10^{-2} per independent synthesis. The bounds also apply only to *this* archive configuration: this code and rate, this placement, this decoder with its 24-read consensus cap, DT4DDS's error model, and the scenarios we ran. They do not extend to DNA storage in general or to an archive with a weaker inner code or a different verification chain. Mechanistically, a false accept would require a wrong payload to pass a 4-symbol RS code, a CRC-16 and an address byte, then the stripe CRC-32 and the file SHA-256. The cryptographic hash is the final safeguard, and no experiment of this size can bound *its* failure rate.

B. Reads consumed versus capacity reserved

The largest effect in this paper is the difference between reads that are *consumed* and capacity that must be *reserved* (Table V). In the baseline digital twin, round-robin with verified stopping consumed 3.49k reads on

TABLE V

READS *consumed* WITH VERIFIED STOPPING VERSUS CAPACITY THAT MUST BE *reserved* WITHOUT IT (DIGITAL TWIN, 300 RETRIEVALS PER ROW). ALL ROWS STOP ON VERIFICATION, SO THE COMPARISON BETWEEN ROWS IS ALLOCATION ONLY; THE LAST COLUMN IS THE BUDGET THE SAME READ PROPORTIONS WOULD NEED WITHOUT STOPPING, WHICH COINCIDES WITH Q99 WHEREVER EVERY RETRIEVAL SUCCEEDS. THOUSANDS OF READS; MEAN OVER ALL ATTEMPTS WITH A CLUSTER-BOOTSTRAP 95% INTERVAL.

Scen.	Allocation	p_{ok}	$\bar{r}_{\text{all}}$ [95% CI]	q95	q99	$B_{0.99}$
Base	Static α=0 (fixed depth)	1.00	3.49 [3.37, 3.60]	4.56	5.15	5.15
	Static α=0.25	1.00	3.45 [3.33, 3.55]	4.60	5.05	5.05
	Static α=0.5	1.00	3.40 [3.29, 3.50]	4.55	4.90	4.90
	Static α=0.75	1.00	3.36 [3.25, 3.46]	4.45	4.80	4.80
	Static α=1	1.00	3.34 [3.24, 3.44]	4.45	4.80	4.80
	Static α=1.25	1.00	3.33 [3.24, 3.43]	4.45	5.00	5.00
	Static α=1.5	1.00	3.33 [3.24, 3.43]	4.40	4.95	4.95
	Static α=2	1.00	3.36 [3.27, 3.44]	4.40	4.90	4.90
	Static α=3	1.00	3.47 [3.38, 3.54]	4.55	5.30	5.30
	Static α=0.5, calibration-estimated	1.00	3.39 [3.29, 3.49]	4.50	4.85	4.85
	Static α=1, calibration-estimated	1.00	3.34 [3.23, 3.43]	4.45	5.00	5.00
	Static α=2, calibration-estimated	1.00	3.39 [3.30, 3.46]	4.40	5.10	5.10
	Round-robin + verified stop	1.00	3.49 [3.37, 3.60]	4.56	5.15	5.15
	Coverage-greedy	1.00	4.02 [3.77, 4.30]	5.80	9.05	9.05
	Pivotal	1.00	5.21 [4.75, 5.70]	8.85	9.60	9.60
	Thompson sampling	1.00	3.75 [3.57, 3.95]	5.21	6.45	6.45
	UCB	1.00	4.02 [3.75, 4.29]	6.05	8.15	8.15
	Learn-then-commit	1.00	5.05 [4.74, 5.31]	6.80	7.65	7.65
	Info-greedy	1.00	18.87 [17.25, 20.31]	31.44	37.45	37.45
	Tail-aware	1.00	3.94 [3.71, 4.19]	5.65	7.15	7.15
Homog.	Static α=0 (fixed depth)	1.00	2.26 [2.23, 2.29]	2.60	2.85	2.85
	Static α=0.5	1.00	2.26 [2.24, 2.29]	2.65	2.85	2.85
	Static α=1	1.00	2.26 [2.24, 2.29]	2.65	2.85	2.85
	Static α=2	1.00	2.26 [2.24, 2.29]	2.65	2.85	2.85
	Static α=0.5, calibration-estimated	1.00	2.26 [2.24, 2.28]	2.65	2.75	2.75
	Static α=1, calibration-estimated	1.00	2.26 [2.23, 2.28]	2.65	2.75	2.75
	Static α=2, calibration-estimated	1.00	2.26 [2.24, 2.28]	2.65	2.75	2.75
	Round-robin + verified stop	1.00	2.26 [2.23, 2.29]	2.60	2.85	2.85
	Coverage-greedy	1.00	2.18 [2.16, 2.20]	2.50	2.70	2.70
	Pivotal	1.00	2.18 [2.16, 2.20]	2.50	2.75	2.75
	Thompson sampling	1.00	2.19 [2.17, 2.22]	2.50	2.70	2.70
	UCB	1.00	2.18 [2.16, 2.20]	2.50	2.75	2.75
	Learn-then-commit	1.00	2.19 [2.17, 2.21]	2.50	2.75	2.75
	Info-greedy	1.00	2.25 [2.23, 2.28]	2.60	2.90	2.90
	Tail-aware	1.00	2.20 [2.18, 2.23]	2.55	2.75	2.75

average (95% CI 3.37–3.60k), while the same uniform read proportions without stopping must reserve 5.15k to succeed on 99% of retrievals, and the best static plan in the family must reserve 4.80k. Stopping converts a 4.80–5.15k reservation into a 3.49k average bill: a 27–32% reduction, depending on which plan the reservation is taken from, and it needs no model of the containers.

Allocation is a separate and much smaller effect. Three static plans are reported in this paper. All three read fixed proportions $w_j \propto e_j^{\alpha}$ *with* the same verified stopping, which is the appropriate basis for a policy-versus-policy comparison, and all three are paired against round-robin on the same 300 syntheses and read orders.

1) *Pre-specified plan* (primary). $\alpha \in \{0, 0.5, 1, 2\}$ was the frozen grid and α=1, which equalises expected coverage, is its best member: 3.343k reads, -4.21% against round-robin's 3.490k (95% CI -4.9 to -3.5%). This is the primary static result.
2) *Extended grid* (secondary sensitivity check). Adding $\alpha \in \{0.25, 0.75, 1.25, 1.5, 3\}$ moves the point estimate to -4.57% at α=1.5 (CI -5.5 to -3.5%). That interval overlaps α=1's almost entirely, so the finer grid does *not* establish a better plan; it shows the family is flat near its optimum (-3.7% to -4.6% for $\alpha \in [0.75, 2]$, decaying to -0.70% at α=3). It is a sensitivity check, not an improved result.
3) *Calibration-estimated plan* (secondary, practical variant). The same α=1 form with the per-container efficiencies e_j measured on the ten *disjoint* calibration replicates, so the plan sees no evaluation data at all: 3.338k reads, -4.35% (CI -5.1 to -3.6%), statistically indistinguishable from the pre-specified hindsight plan. The

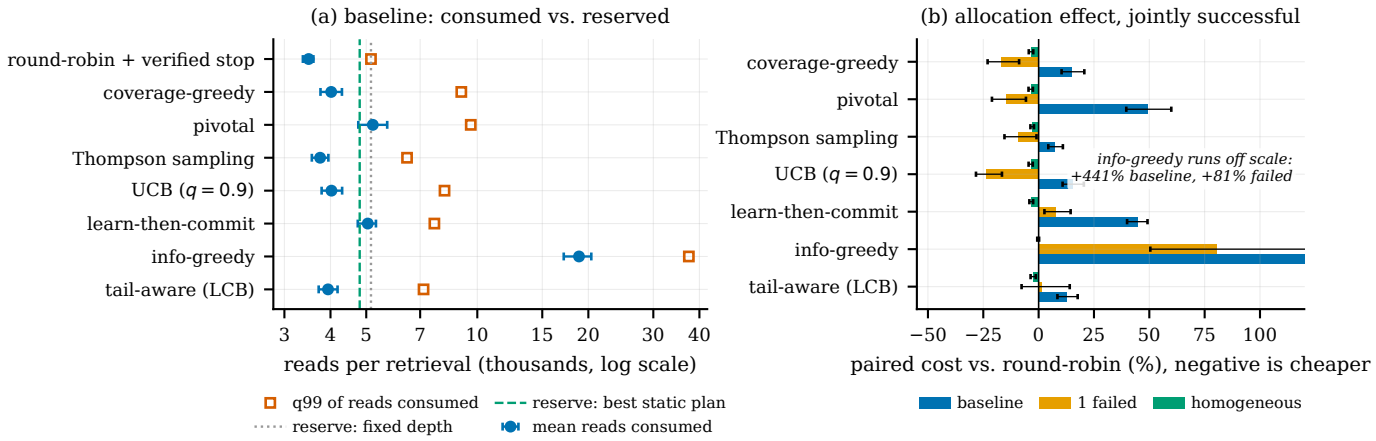


Fig. 2. Digital-twin end-to-end retrieval, baseline unless stated. **(a)** For each policy, the reads a retrieval actually *consumes*: filled circles are the mean over 300 retrievals with a 95% cluster-bootstrap interval over syntheses, open squares the 0.99-quantile of the same retrievals. The two vertical lines are the capacity a plan that does *not* stop must *reserve* to succeed on 99% of retrievals: dotted for uniform fixed depth, dashed for the best static plan. The horizontal distance from a filled circle to a vertical line is the saving verified stopping provides; the distance between circles is the (much smaller) allocation effect. **(b)** Allocation effect alone: each bar is the paired difference in mean reads against round-robin with the same verified stopping, computed on the retrievals in which both policies succeeded, with 95% cluster-bootstrap intervals; negative is cheaper than round-robin. Information-greedy is annotated rather than plotted because it runs off the scale. Scenarios in which no retrieval succeeded (photolithographic, two failed containers) have no paired statistic and are omitted.

static advantage is therefore not an artefact of hindsight.

Against that band of roughly −4%, the model-based rules consume 3.75–18.87k reads, 7–441% *more* than round-robin. On the baseline, fixed proportions measured once and combined with verified stopping are best by about 4%, round-robin with stopping is next, and every online rule we tried is worse. The 27–32% figure above is *not* an allocation result: it compares reads consumed with capacity reserved, and this comparison underlies the recommendation in Section VII.

C. Policy comparison on the digital twin

Table VI and Fig. 2 give the pre-specified comparisons. In the baseline scenario every policy recovered the file in all 300 retrievals. Round-robin with verified stopping used 3.49k reads on average (95% CI 3.37–3.60k). Every model-based rule was costlier on the same syntheses and read orders: Thompson sampling by 7% (4–11%), tail-aware by 13% (9–18%), coverage-greedy by 15% (10–21%), UCB by 15% (11–20%), learn-then-commit by 45% (40–49%), pivotal by 49% (39–60%) and information-greedy by 441% (408–467%). The ordering follows the share of reads spent on the near-dead container: 16% for round-robin, 23% for Thompson sampling, 25–27% for tail-aware, UCB and coverage-greedy, 40–41% for pivotal and learn-then-commit, and 66% for information-greedy. In the homogeneous control (six fresh, independently synthesised containers), where we expected no allocation effect, most model-based rules were 2–4% cheaper than round-robin (coverage-greedy 2.18k vs. 2.26k; information-greedy even at 2.26k): independent syntheses differ enough for mild adaptive gains, consistent with the low-severity end of the sweep below. With DT4DDS's photolithographic synthesis model, no policy and none of the four static plans recovered the file in any of 660 retrievals (eight policies at 10 syntheses × 3 read orders, four static plans, and round-robin re-run at the full 30 × 10 design to confirm the reduced one): every retrieval read the whole 72k-read stream and reported failure, with no wrong output. The two zero-success results differ in kind. For *two failed containers* the limit is arithmetic and follows from the code and placement rather than from our experiments (Section VI-G): exactly $K{=}32$ fragments per stripe remain, and no policy and no read budget can recover the file. For the *photolithographic* scenario the result is empirical and narrower: it holds under that synthesis model with this code, this decoder, a 24-read consensus cap and a 72k-read stream. We did not test deeper streams, a stronger inner code or a lower code rate, and the consensus cap moves the boundary in the neighbouring failure scenario (Section VI-L). This is therefore a limit of *our* archive design and decoder configuration under that synthesis model, not evidence that photolithographic DNA archives are unrecoverable.

TABLE VI

DIGITAL-TWIN END-TO-END RETRIEVAL (DT4DDS; PRIMARY EVIDENCE). PER SCENARIO AND POLICY: SHARE OF RETRIEVALS WHOSE OUTPUT FILE IS BYTE-IDENTICAL TO THE STORED FILE (NO RETRIEVAL RETURNED A WRONG FILE), MEAN READS PER RETRIEVAL (THOUSANDS; CLUSTER-BOOTSTRAP 95% CI OVER SYNTHESES), THE 0.99-QUANTILE Q99 (THE BUDGET TO RESERVE), AND THE PAIRED RELATIVE DIFFERENCE IN MEAN READS AGAINST ROUND-ROBIN WITH VERIFIED STOPPING ON RETRIEVALS WHERE BOTH SUCCEEDED (95% CI). **STATIC PLANS ARE COSTED HERE AS *reservations***: THEY READ FIXED PROPORTIONS UNTIL DECODABLE AND MUST COMMIT THEIR Q99 ON EVERY RETRIEVAL, SO THEIR MEAN COST EQUALS THEIR Q99, AND THEIR PROPORTIONS USE PER-CONTAINER EFFICIENCIES MEASURED IN HINDSIGHT ON THE SAME SCENARIO. TABLE V REPORTS THE SAME PLANS RUN *with* VERIFIED STOPPING, WHERE THEIR CONSUMED MEAN IS FAR LOWER; THE TWO TABLES ANSWER DIFFERENT QUESTIONS AND THEIR STATIC ROWS ARE NOT COMPARABLE.

Scenario	Policy	Success	Mean reads (k) [95% CI]	q99 (k)	vs. round-robin, % [95% CI]
Baseline	Round-robin + verified stop	1.00	3.49 [3.37, 3.60]	5.15	
	Coverage-greedy	1.00	4.02 [3.76, 4.30]	9.05	+15 [+10, +21]
	Pivotal	1.00	5.21 [4.76, 5.70]	9.60	+49 [+40, +60]
	Thompson sampling	1.00	3.75 [3.56, 3.95]	6.45	+7 [+4, +11]
	UCB (q=0.9)	1.00	4.02 [3.78, 4.30]	8.15	+15 [+11, +20]
	Learn-then-commit	1.00	5.05 [4.74, 5.31]	7.65	+45 [+40, +49]
	Info-greedy (exp. design)	1.00	18.87 [17.15, 20.37]	37.45	+441 [+407, +470]
	Tail-aware (LCB)	1.00	3.94 [3.72, 4.18]	7.15	+13 [+8, +18]
	Static, uniform (fixed depth)	1.00	5.15 (= q99)	5.15	
	Static, $w \propto e^{0.5}$	1.00	4.90 (= q99)	4.90	
	Static, $w \propto e$ (equal coverage)	1.00	4.80 (= q99)	4.80	
	Static, $w \propto e^2$	1.00	4.90 (= q99)	4.90	
One container failed	Round-robin + verified stop	0.34	20.31^s	–	
	Coverage-greedy	0.34	16.86^s	–	-17 [-23, -9]
	Pivotal	0.34	17.30^s	–	-15 [-21, -6]
	Thompson sampling	0.34	18.43^s	–	-9 [-15, -1]
	UCB (q=0.9)	0.34	15.49^s	–	-24 [-28, -17]
	Learn-then-commit	0.34	21.91^s	–	+8 [+3, +14]
	Info-greedy (exp. design)	0.34	36.68^s	–	+81 [+50, +130]
	Tail-aware (LCB)	0.34	20.63^s	–	+2 [-8, +14]
	Static, uniform (fixed depth)	0.34	–	–	
	Static, $w \propto e^{0.5}$	0.34	–	–	
	Static, $w \propto e$ (equal coverage)	0.34	–	–	
	Static, $w \propto e^2$	0.34	–	–	
Two containers failed	all 12 policies and static plans	0.00	– (stream exhausted, 72k reads)	–	
Homogeneous (control)	Round-robin + verified stop	1.00	2.26 [2.23, 2.29]	2.85	
	Coverage-greedy	1.00	2.18 [2.16, 2.21]	2.70	-3 [-4, -2]
	Pivotal	1.00	2.18 [2.16, 2.20]	2.75	-4 [-5, -3]
	Thompson sampling	1.00	2.19 [2.17, 2.22]	2.70	-3 [-4, -2]
	UCB (q=0.9)	1.00	2.18 [2.16, 2.20]	2.75	-4 [-5, -3]
	Learn-then-commit	1.00	2.19 [2.17, 2.21]	2.75	-3 [-4, -2]
	Info-greedy (exp. design)	1.00	2.25 [2.23, 2.28]	2.90	-0 [-1, +0]
	Tail-aware (LCB)	1.00	2.20 [2.18, 2.23]	2.75	-2 [-4, -1]
	Static, uniform (fixed depth)	1.00	2.85 (= q99)	2.85	
	Static, $w \propto e^{0.5}$	1.00	2.85 (= q99)	2.85	
	Static, $w \propto e$ (equal coverage)	1.00	2.85 (= q99)	2.85	
	Static, $w \propto e^2$	1.00	2.85 (= q99)	2.85	
Photolithographic synthesis[‡]	all 12 policies and static plans	0.00	– (stream exhausted, 72k reads)	–	

30 independent syntheses of all six containers $\times$ 10 read orders = 300 retrievals per policy and scenario, except [‡]10×3 (protocol deviation, compute). A failed retrieval read the whole stream (72,000 reads) without verifying; ^{s}: mean over successful retrievals only (a mean or quantile including failures is undefined, "–"). When success < 1, success was identical for every policy on every synthesis and read order, so the paired comparison uses the same retrievals for all policies.

D. What the gain after a container failure buys

The one setting in which the online rules won is a container that failed outright, and the size of that gain depends on the denominator (Table VII). With one failed container 34% of retrievals succeeded, and *the success probability was identical for every policy and every static plan*: whether the file can be recovered at all is decided by which fragments survive in the five remaining containers, not by the reading order. Among the 102 retrievals that succeeded, UCB used 15.49k reads against round-robin's 20.31k, 24% fewer. Counting all 300 attempts, including the 66% that read the whole 72k-read stream and failed, UCB used 52.79k against round-robin's 54.43k, a 3%

TABLE VII

ONE FAILED CONTAINER: COST CONDITIONAL ON SUCCESS AND OVER ALL ATTEMPTS (300 RETRIEVALS PER ROW, 72K-READ STREAM). EVERY POLICY HAS THE SAME SUCCESS PROBABILITY, SO THE CONDITIONAL SAVING DOES NOT BUY A BETTER CHANCE OF RECOVERY, AND FAILURES DOMINATE THE UNCONDITIONAL MEAN. THOUSANDS OF READS.

Allocation	p_{ok}	$\bar{r}_{\text{ok}}$	vs. RR	$\bar{r}_{\text{all}}$
Round-robin + verified stop	0.34	20.31	–	54.42
UCB	0.34	15.49	−24	52.79
Static α=1	0.34	15.70	−23	52.86
Static α=0.5	0.34	16.10	−21	52.99
Static α=2	0.34	16.25	−20	53.05
Coverage-greedy	0.34	16.86	−17	53.25
Static α=0 (fixed depth)	0.34	16.91	−17	53.27
Pivotal	0.34	17.30	−15	53.40
Thompson sampling	0.34	18.43	−9	53.79
Tail-aware	0.34	20.63	+2	54.53
Learn-then-commit	0.34	21.91	+8	54.97
Info-greedy	0.34	36.68	+81	59.99

reduction. The two numbers answer different questions: a user who retries until success pays the conditional number, and a user who pays per attempt pays the unconditional one. After an outright failure, posterior rules therefore cut the cost of the retrievals that succeed by up to 24%, without improving the probability of success, and change the cost per attempt by 3%.

E. Why adaptive allocation loses, and what fixes it

The frozen controller models a container by one efficiency multiplier θ_j that scales every fragment's coverage. The near-dead container (two half-lives, 40 PCR cycles, σ=0.5, three copies per strand) instead loses a large fraction of its molecules outright: many of its fragments are never read however long it is sequenced, while the surviving ones are read often. Early batches from it therefore recover fragments at a rate the model explains with a moderate θ_j and extrapolates to the missing fragments, so coverage-based scores keep predicting gains there. The beta-binomial likelihood (ρ_U=0.1) widens the posterior, which is what calibration asked for, and slows the correction further; information-greedy, which reads the container with the largest posterior variance, never brings that variance below its threshold and spends two thirds of its reads there.

Separating the three candidate causes. Three things could be responsible: the container model, the likelihood (ρ_U), and the prior width (s). To separate them we ran the $2\times2\times2$ ablation of Table VIII, and the result was not the one we expected. No single factor is *the* cause. The frozen configuration (θ only, ρ_U=0.1, s=0.5) is the one corner of the cube in which the penalty is large, and relaxing *any* of the three removes most of it. For coverage-greedy, paired against round-robin on the same 300 syntheses and read orders: the frozen corner costs +15.1% (95% CI +10.3 to +20.6%); switching to the binomial likelihood alone gives −4.4% (−6.0 to −2.8%); widening the prior alone gives −0.6% (−2.5 to +1.4%); adding dropout alone gives +1.6% (−0.2 to +3.4%); and combinations reach a floor of about −4.5%, with no combination beating the best single change. Thompson sampling behaves the same way (+7.4%, then −3.5%, +0.9%, +0.7%). The common mechanism is how fast the controller can conclude that a container is bad: the over-dispersed likelihood widens the posterior, a tight prior holds it near θ=1, and a single-multiplier model keeps extrapolating the surviving fragments' rate to the absent ones. Relaxing any one of the three lets the evidence accumulate fast enough, and every rule then falls in the same narrow band a few per cent below round-robin, the band that fixed proportions already occupy (−4.21% for the pre-specified plan, Section VI-B). This band, rather than the +15%, is our estimate of what allocation is worth in this setting.

F. An efficiency-and-dropout container model

We next asked whether a container model that *can* express dropout rescues adaptive allocation. The model uses a joint posterior over (θ_j, d_j) with $d_j \in \{0, 0.05, 0.15, 0.3, 0.5, 0.7, 0.9\}$, so that a fragment still missing after many reads is increasingly attributed to permanent absence rather than to low coverage. It does, with qualifications.

On the baseline, keeping the frozen likelihood and prior and changing only the model, UCB goes from +15.3% to −3.5% against round-robin (95% CI −4.9 to −2.2%), coverage-greedy from +15.1% to +1.6% (−0.2 to +3.4%),

TABLE VIII

ABLATION OF THE THREE MODELLING CHOICES (DIGITAL TWIN; PAIRED MEAN READS AGAINST ROUND-ROBIN ON JOINTLY SUCCESSFUL RETRIEVALS, PER CENT, NEGATIVE IS CHEAPER THAN ROUND-ROBIN). θ: ONE EFFICIENCY MULTIPLIER PER CONTAINER (THE FROZEN MODEL); $\theta{+}d$: EFFICIENCY AND A DROPOUT FRACTION. ρ_U IS THE BETA-BINOMIAL CORRELATION AND s THE PRIOR WIDTH; THE FROZEN CONFIGURATION IS THE FIRST COLUMN. THE PENALTY IS LARGE ONLY IN THAT COLUMN: RELAXING ANY ONE OF THE THREE CHOICES MOVES EVERY RULE INTO A BAND A FEW PER CENT BELOW ROUND-ROBIN. BLANK: NOT RUN. CLUSTER-BOOTSTRAP INTERVALS FOR EVERY CELL ARE IN `RESULTS/R4_ABLATION.CSV`; THOSE THE TEXT RELIES ON ARE QUOTED THERE.

		θ only				θ and dropout d			
Scenario	Policy (ρ_U, s)	0.1, 0.5	0.0, 0.5	0.1, 1.0	0.0, 1.0	0.1, 0.5	0.0, 0.5	0.1, 1.0	0.0, 1.0
baseline	Coverage-greedy	+15	-4	-1	-5	+2	-4	+1	-2
	Thompson sampling	+7	-4	+1	-4	+1	-4	-1	-4
	Tail-aware	+13	-3	-3	-4	+10	-3	-3	-3
	Pivotal	+49	+4	+12	+4	+7	+3	+8	+5
	UCB	+15				-4			
	Learn-then-commit	+45				+13			
	Info-greedy	+441				+612			
fail1	Coverage-greedy	-17	-15	+2		-35	-45	-31	
	Thompson sampling	-9	-20	-7		-36	-43	-29	
	Tail-aware	+2	-25	-23		-39	-42	-22	
	Pivotal	-15	-8	+16		-34	-45	-30	
	UCB	-24				-35			
	Learn-then-commit	+8							
	Info-greedy	+81							
homog	Coverage-greedy	-3				-3			
	Thompson sampling	-3				-3			
	Tail-aware	-2				-2			
	Pivotal	-4				-4			
	UCB	-4				-4			
	Learn-then-commit	-3				-3			
	Info-greedy	-0				-0			
sweep-cov3	Coverage-greedy	+15				+3			
	Thompson sampling	+7				+1			
	Tail-aware	+15				+11			
	Pivotal					+10			
	UCB					-3			
	Learn-then-commit					+15			
	Info-greedy					+678			
sweep-cov1	Coverage-greedy	+13				-5			
	Thompson sampling	+8				-5			
	Tail-aware	+7				+2			
	Pivotal					-0			
	UCB					-11			
	Learn-then-commit					+10			
	Info-greedy					+679			

Thompson sampling from $+7.4\%$ to $+0.7\%$, pivotal from $+49.3\%$ to $+7.4\%$ and learn-then-commit from $+44.6\%$ to $+12.7\%$. The mechanism is visible in the read shares: coverage-greedy's share of the near-dead container falls from 0.27 to 0.20 and UCB's from 0.27 to 0.17, against round-robin's 0.16. Information-greedy gets worse ($+441\%$ to $+612\%$), as expected: a richer model gives a pure-information rule more uncertainty to chase. In the homogeneous control, where there is no dropout to detect, the two models agree within 0.5% on every rule, so the second parameter costs nothing when it is not needed. The clearest benefits are in the scenarios the one-parameter model misreads most. After an outright failure, keeping the frozen likelihood, the conditional saving grows from -17.0% to -35.0% for coverage-greedy, from -9.2% to -36.2% for Thompson sampling and from $+1.6\%$ to -38.5% for tail-aware, with the success probability unchanged at 0.34; adding the binomial likelihood as well takes all four rules to between -42% and -45%. Under partial physical loss and common-cause degradation the two-parameter model turns penalties of $+77\%$ and $+47\%$ into savings of -6% and -34% (Section VI-L).

Three qualifications apply. First, as the ablation shows, the dropout model is not uniquely responsible: the binomial likelihood alone does as much on the baseline. Second, the tail does not move for any *adaptive* rule.

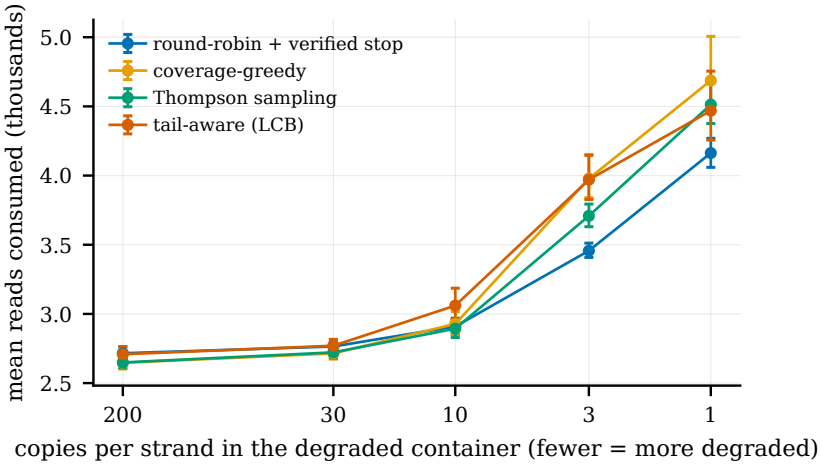


Fig. 3. Severity sweep: mean reads consumed as the degraded container's copy number falls from 200 to 1 (300 retrievals per point and policy, 95% cluster-bootstrap intervals; the axis runs from mild to severe degradation). Every retrieval at every level returned the correct file, so the success probability is 1.00 throughout and is not plotted. Adaptive rules are cheaper than round-robin only at the mild end.

The paired q99 difference for UCB with the dropout model is -450 reads with a 95% interval of $-1{,}147$ to $+150$, so even the best rule is not resolved as better than round-robin at q99, whereas a hindsight static plan is (Section VI-H). Third, the size of the effect depends on the scenario. On the baseline the best allocation result is a 3.5–4.5% mean saving, against the 27–32% that verified stopping delivers with no model. Allocation reaches 6–45% only in the dropout-dominated scenarios (partial loss, common-cause degradation, a failed container), where it repairs a problem the frozen model created. The negative result is therefore not that adaptive allocation does not work. Rather, a controller that is slow to recognise a bad container makes allocation harmful; several independent modelling choices cause that slowness; and fixing any of them gains a few per cent, not the bulk of the saving.

G. *Failed containers and severity sweep*

With one failed container (the fresh one, which returns only non-informative reads), 34% of retrievals succeeded under every policy, with identical success on every synthesis and read order (Section VI-D); recoverability is set by the archive's redundancy, not by the policy. On the 102 retrievals where all policies succeeded, adaptive allocation helped, because the failed container is easy to identify and stops being read: UCB used 24% (17–28%) fewer reads than round-robin, coverage-greedy 17% (8–23%), pivotal 15% (6–21%) and Thompson sampling 9% (2–15%); tail-aware was even ($+2$%, -8 to $+14$%), learn-then-commit 8% (2–14%) and information-greedy 81% (49–133%) costlier. With two failed containers no policy recovered the file; every retrieval read all 72,000 reads and reported failure. Both results are limits of the *archive design*, not of the policies. With round-robin placement of $(32, 48)$ stripes over six containers, each failed container removes 8 of a stripe's 48 fragments, half of its redundancy of 16. After one failure a stripe tolerates only 8 further lost strands: 18 of the 30 syntheses were unrecoverable under every read order, 7 always recoverable, and 5 depended on the read order (consensus uses at most 24 reads per cluster); after two failures exactly $K{=}32$ fragments remain, so a single strand lost by synthesis, aging or PCR in any stripe makes the file unrecoverable. Surviving more failures requires more redundancy or a different placement; no read-allocation policy can compensate.

The severity sweep (Fig. 3) varies only the near-dead container's copies per strand. At 200 and 30 copies, coverage-greedy and Thompson sampling are 2–3% cheaper than round-robin (2.65k vs. 2.72k at 200 copies) and tail-aware is even; at 10 copies the policies are within ± 1% except tail-aware ($+5$%); at 3 and 1 copies all adaptive rules are costlier (at one copy: coverage-greedy 4.69k, Thompson 4.51k, tail-aware 4.47k vs. 4.16k). Success was 1.00 throughout. The point at 3 copies reproduces the baseline design with other seeds (round-robin 3.46k vs. 3.49k). Adaptive allocation therefore helps only while the model's single multiplier is a fair description of the worst container.

TABLE IX

TAIL METRICS WITH CLUSTER-BOOTSTRAP 95% INTERVALS OVER SYNTHESIS REPLICATES (BASELINE). EXTREME QUANTILES ESTIMATED FROM A FEW HUNDRED RETRIEVALS ARE IMPRECISE, AND THE INTERVALS OF MOST POLICIES OVERLAP; THE ORDERING OF THE MEANS DOES NOT CARRY OVER TO Q99. THOUSANDS OF READS.

Design	Allocation	$\bar{r}_{\text{all}}$	q95 [95% CI]	q99 [95% CI]
30 syntheses × 10 orders	Static α=1	3.34	4.45 [4.01, 4.55]	4.80 [4.60, 5.25]
	UCB + dropout model	3.37	4.25 [4.05, 4.45]	4.70 [4.40, 5.05]
	Static α=0 (fixed depth)	3.49	4.56 [4.30, 4.75]	5.15 [4.75, 5.65]
	Round-robin + verified stop	3.49	4.56 [4.30, 4.75]	5.15 [4.75, 5.65]
	Thompson sampling + dropout model	3.52	4.60 [4.20, 5.00]	5.45 [4.65, 5.75]
	Coverage-greedy + dropout model	3.55	4.55 [4.30, 5.05]	5.10 [4.65, 5.45]
	Pivotal + dropout model	3.75	4.80 [4.60, 5.05]	5.30 [4.95, 5.85]
	Thompson sampling	3.75	5.21 [4.70, 6.10]	6.45 [5.35, 7.19]
	Tail-aware + dropout model	3.82	5.10 [4.80, 5.55]	6.00 [5.35, 6.60]
	Learn-then-commit + dropout model	3.93	5.15 [4.85, 5.61]	5.85 [5.20, 6.55]
	Tail-aware	3.94	5.65 [5.15, 6.19]	7.15 [5.75, 7.75]
	Coverage-greedy	4.02	5.80 [5.25, 8.05]	9.05 [5.80, 9.50]
	UCB	4.02	6.05 [5.20, 7.50]	8.15 [6.15, 8.60]
	Learn-then-commit	5.05	6.80 [6.35, 7.20]	7.65 [6.95, 8.10]
	Pivotal	5.21	8.85 [8.10, 9.25]	9.60 [9.10, 9.90]
	Info-greedy	18.87	31.44 [27.60, 34.22]	37.45 [33.70, 37.51]
	Info-greedy + dropout model	24.84	37.20 [36.00, 37.30]	37.50 [37.20, 37.51]
100 syntheses × 3 orders	Static α=1	3.33	4.35 [4.05, 4.60]	4.70 [4.55, 4.75]
	Thompson sampling + dropout model	3.48	4.40 [4.25, 4.60]	5.10 [4.60, 5.46]
	Static α=0 (fixed depth)	3.50	4.56 [4.35, 4.76]	5.15 [4.85, 5.20]
	Round-robin + verified stop	3.50	4.56 [4.30, 4.76]	5.15 [4.85, 5.20]
	Coverage-greedy + dropout model	3.52	4.36 [4.20, 4.55]	5.05 [4.55, 5.11]
	Thompson sampling	3.75	5.01 [4.75, 5.55]	6.10 [5.40, 7.16]
	Tail-aware	3.94	5.50 [5.20, 5.75]	6.90 [5.71, 7.16]
	Coverage-greedy	4.00	5.75 [5.25, 6.05]	8.05 [5.85, 9.06]
	UCB	4.04	5.96 [5.21, 6.85]	7.50 [6.15, 8.16]
	Learn-then-commit	5.13	7.16 [6.05, 7.51]	7.70 [7.20, 8.10]
	Pivotal	5.24	9.23 [7.25, 9.46]	9.63 [9.15, 9.85]
	Info-greedy	19.35	32.53 [26.80, 35.57]	36.02 [32.15, 38.25]

H. Tail metrics and the static family

No adaptive rule lowered the 0.99-quantile *reliably*; a tuned static plan did. A q99 estimated from 300 retrievals is driven by its three most expensive observations, which come from a few synthesis replicates. We therefore report cluster-bootstrap intervals over syntheses and re-ran the baseline with 100 independent syntheses × 3 read orders (Table IX). We distinguish a *numerical* reduction in the estimated q99 from a reduction that survives its interval.

Numerically, three adaptive configurations are below round-robin. With the dropout model in the frozen design, UCB has q99 4.70k against round-robin's 5.15k (-8.7%) and coverage-greedy 5.10k (-1.0%); in the 100-synthesis design, coverage-greedy with dropout reaches 4.69k (-9.0%) and Thompson sampling with dropout 4.75k (-7.8%). None of these survives: the paired differences are -450 reads (95% CI $-1{,}001$ to $+242$), -52 (-648 to $+350$), -264 (-646 to $+100$) and -204 (-546 to $+218$), every interval containing zero. Every *other* adaptive rule is significantly *worse* at q99, in both designs: coverage-greedy under the frozen model $+2{,}902$ reads (95% CI $+800$ to $+3{,}904$), Thompson sampling $+953$ ($+300$ to $+2{,}009$), UCB $+2{,}352$ ($+1{,}151$ to $+3{,}003$), tail-aware $+1{,}752$ ($+650$ to $+2{,}149$), learn-then-commit $+2{,}516$, pivotal $+4{,}444$ and information-greedy $+30{,}842$. The one reduction that does survive is not adaptive at all: the pre-specified fixed proportions reach q99 4.70k against round-robin's 5.15k, a paired -450 reads (95% CI -550 to -202), an interval excluding zero. Adaptive rules therefore did not reliably lower the tail, although we cannot rule out that the best of them lowers it by up to about 9%; fixed proportions chosen in advance did lower it. The tail-aware rule, designed to target the bottleneck stripes, is significantly worse than round-robin at q99 in every design we ran, so this heuristic did not achieve its aim. We report q95 beside q99 because it is the better-determined tail statistic at this sample size.

The same caution applies to the static family. Its members are $w_j \propto e_j^{\alpha}$, and the frozen protocol tested only $\alpha \in \{0, 0.5, 1, 2\}$, so "best static plan" means the best of those four, not the optimal fixed allocation; the extended grid and the calibration-estimated variant, and what each of the three numbers means, are defined in Section VI-B

TABLE X

RE-RUN OF THE ρ_U SELECTION WITH THE PRIOR ACTUALLY USED IN THE POLICY RUNS (s=0.5) AND AN EXTENDED GRID: COVERAGE OF THE 90% CREDIBLE INTERVAL OF θ_j^*, OVERALL AND PER CONTAINER. NO VALUE IN THE GRID REACHES 0.90 OVERALL, AND NO VALUE COVERS THE NEAR-DEAD CONTAINER.

Split	ρ_U	All	fresh	aged 1 hl	aged 2 hl, 10 cop.	30 cyc.	45 cyc., 20 cop.	near-dead
select (reps 1000-1009)	0.0	0.43	0.80	0.60	0.00	0.75	0.45	0.00
	0.003	0.64	1.00	0.85	0.25	0.95	0.80	0.00
	0.01	0.74	1.00	0.90	0.55	1.00	1.00	0.00
	0.03	0.81	1.00	0.95	0.90	1.00	1.00	0.00
	0.1	0.81	0.95	0.90	1.00	1.00	1.00	0.00
	0.2	0.80	0.90	0.90	1.00	1.00	1.00	0.00
	0.3	0.80	0.90	0.90	1.00	1.00	1.00	0.00
	0.5	0.60	0.50	0.60	1.00	0.50	1.00	0.00
validate (reps 0-29)	0.0	0.54	0.78	0.88	0.23	0.78	0.52	0.02
	0.003	0.67	0.97	1.00	0.37	0.97	0.68	0.03
	0.01	0.76	1.00	1.00	0.62	1.00	0.93	0.03
	0.03	0.83	1.00	1.00	0.95	1.00	1.00	0.03
	0.1	0.82	0.92	1.00	1.00	0.97	1.00	0.03
	0.2	0.80	0.87	0.97	1.00	0.93	1.00	0.03
	0.3	0.80	0.87	0.97	1.00	0.93	1.00	0.03
	0.5	0.59	0.52	0.55	1.00	0.48	0.98	0.03

and listed in Table V. A user who must reserve capacity gains nothing from stopping unless unused capacity is refunded; for that user the relevant comparison is between reservations, and a tuned static plan is the better choice. In the homogeneous control the static plans must reserve 2.85k against a round-robin consumed mean of 2.26k. With a failed container no static plan can be sized at all: it either reads the whole stream or fails, so its reservation is the full 72k reads.

I. Calibration of the posterior, and a negative forecast result

We consider two calibration questions separately: whether the posterior over θ_j is calibrated, and whether the completion forecast is. Neither is, and the first failure has a specific cause.

Posterior intervals. On the ten selection replicates, 90% credible-interval coverage of θ^* rose with ρ_U from 0.45 (binomial) through 0.63, 0.73 and 0.80 to 0.87 at ρ_U=0.1, the edge of the pre-specified grid, which we kept and selected; on the 30 held-out replicates it gave 0.87 overall (0.84 at the earlier and 0.89 at the later read count), 1.00 for five containers and 0.20 for the near-dead one. That selection has two defects: it used prior sd s=1 while every policy run uses s=0.5, and the selected value sat at the grid boundary. A secondary analysis repeats the selection with s=0.5 and an extended grid $\{0, 0.003, 0.01, 0.03, 0.1, 0.2, 0.3, 0.5\}$ (Table X). *Every policy run in this paper, including all primary results and all secondary scenarios, uses the originally selected* ρ_U=0.1; the re-selection is a *sensitivity experiment* that re-runs the selection procedure only, changes no policy run and replaces no reported result. The original selection and its two defects are documented in Appendix B. The re-selection corrects the original in two ways. First, under the prior actually used the best value is ρ_U=0.03 (selection coverage 0.81, held out 0.83) rather than 0.1 (0.81 and 0.82), so the frozen default is not the best member of its own family, although the difference is within noise. Second, extending the grid does not rescue calibration: coverage flattens near 0.80 for $\rho_U \in [0.03, 0.3]$ and collapses to 0.60 at 0.5, so no value in this likelihood family reaches the 0.90 target. The reason is visible per container: at every ρ_U the five healthy containers are covered 0.92–1.00 of the time and the near-dead container 0.00–0.03 of the time. Widening a likelihood cannot fix a container whose behaviour the model cannot express, which is the same misspecification that makes allocation lose (Section VI-E). The controller is therefore not calibrated: its posterior is reliable for healthy containers and wrong for near-dead ones, which is the case that matters. No primary conclusion changes as a result. The re-selected value, ρ_U=0.03, differs from the frozen 0.1 by less than the noise in the selection statistic, and the ablation of Table VIII already covers the likelihood axis: the allocation penalty depends on ρ_U as one of three interacting choices, and we report this as a finding. The primary experiments were not re-tuned to the new value, so that the protocol remains as frozen.

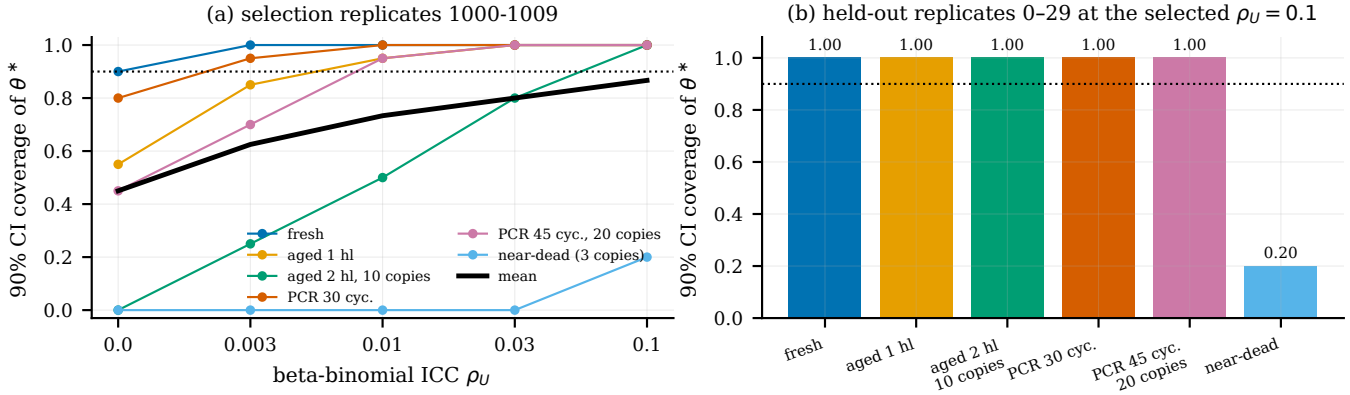


Fig. 4. Is the posterior over container efficiency calibrated? Each point or bar is the fraction of cases in which the 90% credible interval for θ_j contained the target θ_j^*; a calibrated posterior would sit on the dotted line at 0.90. **(a)** Selection replicates 1000–1009, one curve per container, against the beta-binomial correlation ρ_U; the thick black curve is the mean over containers, which is what the selection rule maximised. **(b)** Held-out replicates 0–29 at the selected ρ_U=0.1, per container. Five containers are covered perfectly and the near-dead one is covered in 0.20 of cases at this ρ_U (0.00–0.03 across the extended grid of Table X): no value of ρ_U calibrates the container that matters.

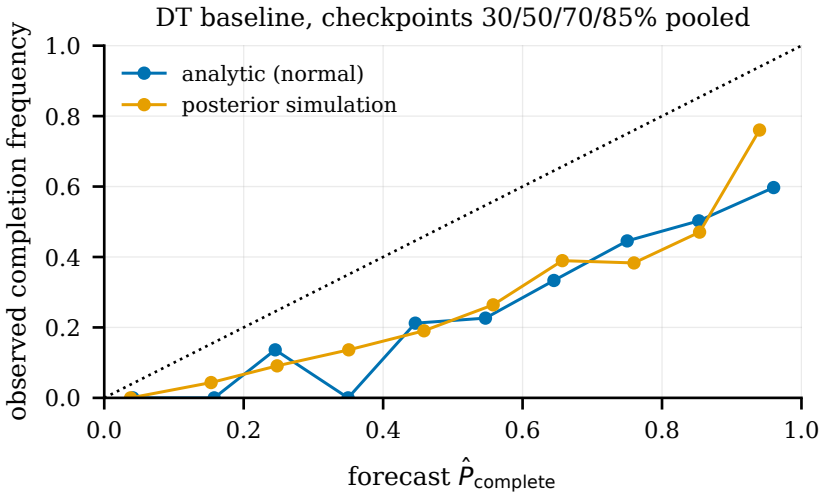


Fig. 5. Reliability of the completion forecast on the digital twin: forecast probability of finishing within the remaining budget against the observed frequency of finishing, pooled over checkpoints at 30/50/70/85% of the budget (10 equal-width bins, bins with fewer than five checkpoints dropped). A calibrated forecast would lie on the diagonal; both curves lie below it, so both are over-confident, and both have a worse Brier score than a constant forecast at the base rate.

Completion forecast (negative, secondary). No policy depends on the forecast; we report it because it was pre-specified and its failure is informative. On 1,127 checkpoints from 300 baseline retrievals with a budget of 3.75k reads (the median coverage-greedy cost), of which 52% finished within the budget, both forecasts were over-confident (mean forecast 0.86 against 0.52 observed at 30% of the budget). The posterior-simulation forecast beat the analytic one on every metric (ECE 0.29 vs. 0.34; Brier 0.30, 95% CI 0.27–0.34, vs. 0.34, 0.29–0.38; AUC 0.73, 0.69–0.78, vs. 0.69, 0.65–0.73), yet both Brier scores are worse than the 0.25 of a constant forecast at the observed base rate, so neither beats climatology. They rank retrievals usefully only late in a retrieval (AUC 0.82 at 85% of the budget). Futility stopping, which would rely on them, is disabled in every run, and the forecasts should not be used as probabilities.

J. Primary comparison under the re-selected likelihood

The primary adaptive-allocation comparison runs at ρ_U=0.1, a value chosen by a procedure that used the wrong prior width (Section VI-I); the corrected procedure picks ρ_U=0.03. We therefore re-ran the *complete* primary

TABLE XI

THE COMPLETE PRIMARY COMPARISON REPEATED UNDER EACH LIKELIHOOD SETTING: THE FROZEN ρ_U=0.1, THE VALUE THE CORRECTED SELECTION PICKS (ρ_U=0.03, SECTION VI-I) AND THE BINOMIAL (ρ_U=0). ALL THREE USE THE FULL DESIGN (30 SYNTHESES × 10 READ ORDERS PER POLICY) AND THE FROZEN PRIOR WIDTH s=0.5. ENTRIES ARE THE PAIRED DIFFERENCE IN MEAN READS AGAINST ROUND-ROBIN ON JOINTLY SUCCESSFUL RETRIEVALS, PER CENT, WITH CLUSTER-BOOTSTRAP 95% INTERVALS; NEGATIVE IS CHEAPER THAN ROUND-ROBIN. ROUND-ROBIN NEVER EVALUATES THE POSTERIOR, SO IT IS BIT-IDENTICAL ACROSS THE THREE SETTINGS AND IS THE COMMON REFERENCE. SUCCESS PROBABILITY IS UNCHANGED BY ρ_U IN EVERY SCENARIO.

Scenario	Policy	ρ_U=0.1 (frozen)	ρ_U=0.03 (re-selected)	ρ_U=0 (binomial)
baseline	Coverage-greedy	+15 [+10, +21]	-0 [-2, +2]	-4 [-6, -3]
	Thompson sampling	+7 [+4, +11]	+0 [-2, +2]	-4 [-5, -2]
	UCB	+15 [+11, +20]	+2 [-0, +4]	
	Tail-aware	+13 [+9, +17]	-1 [-3, +1]	-3 [-5, -1]
	Pivotal	+49 [+39, +60]	+12 [+9, +16]	+4 [+2, +7]
	Learn-then-commit	+45 [+40, +49]	+35 [+30, +40]	
	Info-greedy	+441 [+408, +471]	+576 [+531, +617]	
homog	Coverage-greedy	-3 [-5, -2]	-4 [-5, -3]	
	Thompson sampling	-3 [-4, -2]	-3 [-4, -2]	
	UCB	-4 [-5, -3]	-4 [-5, -3]	
	Tail-aware	-2 [-4, -1]	-3 [-4, -2]	
	Pivotal	-4 [-5, -3]	-4 [-5, -3]	
	Learn-then-commit	-3 [-4, -2]	-3 [-4, -3]	
	Info-greedy	-0 [-1, +0]	-0 [-1, +0]	
fail1	Coverage-greedy	-17 [-23, -9]	-9 [-13, -3]	-15 [-20, -10]
	Thompson sampling	-9 [-15, -1]	-11 [-15, -5]	-20 [-23, -17]
	UCB	-24 [-28, -17]	-2 [-9, +6]	
	Tail-aware	+2 [-7, +13]	-17 [-22, -11]	-25 [-29, -21]
	Pivotal	-15 [-21, -6]	+1 [-6, +10]	-8 [-14, -2]
	Learn-then-commit	+8 [+3, +14]	-18 [-21, -14]	
	Info-greedy	+81 [+50, +128]	+80 [+49, +128]	

comparison at the re-selected value: all eight policies, all three primary scenarios, the full 30 syntheses × 10 read orders, and everything else — seeds, prior width s=0.5, container model, placement, decoder — held at its frozen setting. Table XI places it beside the frozen runs and the binomial ablation. Round-robin never evaluates the posterior, so its retrievals are bit-identical across all three settings and serve as a common reference.

The likelihood setting changes part of the allocation result. On the baseline, four of the seven model-based rules move from clearly costlier than round-robin to indistinguishable from it: coverage-greedy from +15.1% to −0.4% (95% CI −2.5 to +1.7%), Thompson sampling from +7.4% to +0.3% (−1.7 to +2.4%), UCB from +15.3% to +1.5% (−0.5 to +3.8%) and tail-aware from +12.9% to −1.2% (−3.2 to +0.6%), every interval containing zero. Three rules stay clearly worse at any setting: pivotal (+12.2%), learn-then-commit (+34.9%) and information-greedy (+576%, worse than at the frozen value). In the homogeneous control the likelihood is irrelevant, as it should be when there is no near-dead container to misread: every rule sits at −2.8% to −3.7% under both settings. After an outright failure the picture is mixed rather than uniformly better: coverage-greedy keeps less of its gain (−8.7% against −17.0%) and UCB most of all loses it (−2.3%, not significant, against −23.7%), while tail-aware (+1.6% to −16.7%) and learn-then-commit (+7.9% to −17.8%) gain. No single ρ_U is best everywhere.

Three consequences follow. (i) The strong form of the negative result, that every model-based rule read 7–441% more than round-robin, holds for the *pre-registered* configuration but not for the corrected one. Under ρ_U=0.03 the coverage family is *no better* than round-robin rather than harmful. (ii) The weaker form holds in every setting we tested, and it is the one we rely on: *no adaptive rule beat round-robin by more than a few per cent in any (scenario, likelihood) combination, while three of the seven were clearly worse in most of them.* (iii) The main conclusion is unaffected and, if anything, reinforced: at the corrected likelihood, adaptive allocation on the baseline gains essentially nothing, against the 27–32% that verified stopping delivers. We keep ρ_U=0.1 as the primary configuration because it was pre-registered and re-tuning after seeing outcomes would break the frozen protocol, but every statement about the *size* of the allocation penalty is reported with the likelihood it depends on.

TABLE XII

SECONDARY EVIDENCE: SEMI-EMPIRICAL REPLAY OF REAL ILLUMINA READ STREAMS (ENA PRJEB65931), GENERIC PRIOR, NO EVALUATION-SPLIT STATISTICS. MEAN READS (THOUSANDS) AND Q99; Δ: MEAN MINUS THE HINDSIGHT-TUNED STATIC BUDGET (PAIRED BOOTSTRAP 95% CI). DECODABILITY IS AN ORACLE EVENT HERE (CONSENSUS EQUALS THE DESIGNED STRAND), SO THIS IS *not* EXACT RETRIEVAL.

Pool	Policy	Mean	q99	Δ vs. static [95% CI]
Twist	Static, uniform (hindsight budget)	68.1	68.1	
	Static, hindsight-tuned	65.7	65.7	
	Round-robin + verified stop	58.6	68.4	-7.1 [-17.4, -6.5]
	Coverage-greedy	56.7	66.0	-9.0 [-19.3, -8.5]
	Pivotal	56.6	67.2	-9.1 [-19.4, -8.6]
	Thompson sampling	56.9	66.0	-8.9 [-19.1, -8.4]
	UCB	56.7	66.6	-9.0 [-19.2, -8.5]
	Learn-then-commit	57.2	66.6	-8.5 [-18.8, -8.0]
	Info-greedy	56.7	66.0	-9.0 [-19.3, -8.5]
	Tail-aware (LCB)	56.7	65.6	-9.0 [-19.3, -8.5]
	Coverage-greedy, leak-free calibrated	56.6	67.6	-9.1 [-19.4, -8.6]
Genscript	Static, uniform (hindsight budget)	852	852	
	Static, hindsight-tuned	631	631	
	Round-robin + verified stop	637	853	+6.0 [-57.4, +15.1]
	Coverage-greedy	639	820	+8.2 [-54.9, +17.2]
	Pivotal	639	820	+8.2 [-54.9, +17.2]
	Thompson sampling	654	831	+23.4 [-39.4, +31.9]
	UCB	637	830	+6.0 [-57.8, +14.9]
	Learn-then-commit	651	843	+19.7 [-42.8, +29.2]
	Info-greedy	not completed (Appendix B)		
	Tail-aware (LCB)	644	827	+12.9 [-51.0, +21.6]
	Coverage-greedy, leak-free calibrated	533	676	-97.9 [-161.6, -90.0]

K. Secondary evidence: real read streams

Table XII gives the replay with the generic prior. On the Twist pool, round-robin with verified stopping used 58.6k reads on average against the 65.7k budget of the hindsight-tuned static plan (-11%), and the model-based rules used 56.6–57.2k, 2–3% fewer than round-robin; coverage-greedy was 9.0k (95% CI 8.5–19.3k) below the hindsight static budget. Information-greedy reduced to coverage-greedy after the first probes (identical results), and the calibrated belief changed coverage-greedy by -0.2%. As on the digital twin, no *adaptive* rule lowered q99 materially: 65.6–67.2k for the model-based rules against the 65.7k static budget. Stopping supplies most of the gain (about 7.1k of 9.0k reads). On the Genscript pool, whose PCR-30-cycle container is near-dead, verified stopping with round-robin used 637k reads against 852k for fixed depth (-25%), but 1% more than the 631k of the hindsight-tuned static plan, which knows in advance to give that container 2.7% of its reads (paired difference +6.0k, 95% CI -57.4 to $+15.1$k). Every model-based rule with the generic prior was 0–3% costlier than round-robin (UCB 637k, coverage-greedy and pivotal 639k, tail-aware 644k, learn-then-commit 651k, Thompson sampling 654k), and, as on the digital twin, each gave the near-dead container more reads than round-robin did (19–20% against 17%). The exception is the calibrated belief, fitted on calibration reads of the same containers that are disjoint from the evaluation reads: coverage-greedy with it gave the near-dead container 3.4% of its reads and used 533k (-16% vs. round-robin; -97.9k, 95% CI -161.6 to -90.0k, vs. the hindsight static budget). Allocation therefore pays when the weak container is identified *before* retrieval, not when the controller must learn it online. q99 was 820–853k for the generic-prior rules and 676k with the calibrated belief, against the 631k static budget. The information-greedy run on this pool did not complete in time and is treated as censored (Appendix B).

What the calibrated prior costs. That 16% is the largest allocation effect in this paper, and it is not free: the calibrated belief is fitted on the first 300,000 reads of each of the six runs, 1.8M reads in total, which is 2.8 times what one retrieval costs. Table XIII charges them. Beliefs built from 1k, 3k or 10k reads per container (counts only, generic consensus curves) save nothing measurable — 640k, 638k and 636k reads against round-robin's 637k — and 30k per container is actually worse (658k), because a half-informative estimate of a container's abundance distribution misleads the allocator more than no estimate at all. Only at 100k reads per container (577k, 600k calibration reads) and at the full 300k (533k, 1.8M calibration reads) does the saving appear. Dividing the

TABLE XIII

WHAT THE CALIBRATED PRIOR COSTS (GENSCRIPT REPLAY, COVERAGE-GREEDY). CALIBRATION READS ARE CHARGED EXPLICITLY: THEY ARE THE FIRST READS OF THE SAME SIX SEQUENCING RUNS, DISJOINT FROM THE EVALUATION READS. BREAK-EVEN IS THE NUMBER OF RETRIEVALS OVER WHICH THE CALIBRATION READS ARE REPAID BY THE PER-RETRIEVAL SAVING AGAINST ROUND-ROBIN. THOUSANDS OF READS.

Belief	Calib./cont.	Calib. total	Retrieval	Saving vs. RR	Break-even
generic	0.00	0.00	639.29	-2.35	–
counts	1.00	6.00	640.34	-3.40	–
counts	3.00	18.00	637.74	-0.80	–
counts	10.00	60.00	635.86	1.08	55.4
counts	30.00	180.00	658.18	-21.24	–
counts	100.00	600.00	576.89	60.05	10.0
full	300.00	1800.00	533.18	103.76	17.3

TABLE XIV

SENSITIVITY TO THE CONSENSUS CAP, THE NUMBER OF READS PER CLUSTER USED TO FORM A CONSENSUS (FROZEN VALUE 24). ON THE BASELINE THE CAP IS NOT BINDING: CLUSTERS RARELY REACH A DOZEN READS BEFORE THE FRAGMENT DECODES. THOUSANDS OF READS.

Scen.	Allocation	Cap	p_{ok}	$\bar{r}_{\text{ok}}$	q99
baseline	Coverage-greedy	12	1.00	4.13	9.13
		24	1.00	4.14	9.13
		48	1.00	4.13	9.13
baseline	Coverage-greedy + dropout model	12	1.00	3.56	5.18
		24	1.00	3.57	5.18
		48	1.00	3.57	5.18
baseline	Round-robin + verified stop	12	1.00	3.53	5.18
		24	1.00	3.53	5.18
		48	1.00	3.53	5.18
fail1	Round-robin + verified stop	12	0.31	20.58	72.00
		24	0.34	20.31	72.00
		48	0.36	21.19	72.00

calibration reads by the per-retrieval saving gives the break-even point: about 10 retrievals for the 100k belief and 17 for the full one. Calibration therefore pays only for an archive that will be read repeatedly, and for a single retrieval it costs several times what it saves. A saving that is large per retrieval thus becomes small or negative once the information it depends on is charged for.

L. Robustness

Prior assumptions. The controller starts from fixed generic assumptions, so we varied them one at a time on the baseline (30 syntheses $\times$ 3 read orders each): the prior probability of outright failure (10^{-2}, 10^{-1} against the frozen 10^{-3}), the prior dropout (0, 5% against 0.5%), the negative-binomial shape (3, 30 against 10), the θ grid (40 points over $[0.001, 10]$ and 10 points over $[0.01, 5]$ against 25 over $[0.01, 5]$), the prior width (s=0.25, 1.0 against 0.5) and the consensus-success curve (scaled by 0.5 and 2.0). Under the one-parameter model the penalty is insensitive to these settings: coverage-greedy stays between $+10.9\%$ and $+20.6\%$ against round-robin in every setting except the tighter prior s=0.25, where it rises to $+56.4\%$, because a prior that holds θ near 1 is the most harmful choice here. Changing the failure prior by two orders of magnitude has no effect ($+17.2\%$ either way), because the failure atom is not what is mismodelled. Under the two-parameter model the same sweep stays between -0.3% and $+3.4\%$. Neither the negative result nor its repair is therefore an artefact of the prior.

Decoder configuration. The consensus cap (reads per cluster used to form a consensus, frozen at 24) is not binding in the baseline: caps of 12, 24 and 48 give the same means to within 10 reads and the same q99 (Table XIV), because clusters there rarely accumulate more than a dozen reads before the fragment decodes. Where a stream is read to exhaustion, however, the cap does bind and shifts what we described above as an archive-design limit. With one failed container, round-robin recovers the file in 31.3% of retrievals at a cap of 12, 34.0% at 24 and 35.7%

TABLE XV

FAILURE MODES BEYOND TOTAL LOSS (20 SYNTHESES × 3 READ ORDERS). PAIRED AGAINST ROUND-ROBIN ON JOINTLY SUCCESSFUL RETRIEVALS. THOUSANDS OF READS.

Setting	Allocation	n	p_{ok}	$\bar{r}_{\text{ok}}$	vs. RR	q99
partial-c0-50	Round-robin + verified stop	60	1.00	6.66	–	13.31
	Coverage-greedy	60	1.00	11.76	+77	21.44
	Coverage-greedy + dropout model	60	1.00	6.26	−6	10.09
	Tail-aware	60	1.00	12.79	+92	31.04
	Thompson sampling	60	1.00	10.24	+54	23.50
	Thompson sampling + dropout model	60	1.00	6.73	+1	12.35
	UCB	60	1.00	10.64	+60	17.55
	Static α=0 (fixed depth)	60	1.00	6.66	+0	13.31
	Static α=1	60	1.00	5.76	−14	11.07
partial-c0c3-25	Round-robin + verified stop	60	0.95	6.45	–	72.00
	Coverage-greedy	60	0.95	10.19	+58	72.00
	Coverage-greedy + dropout model	60	0.95	6.37	−1	72.00
	Tail-aware	60	0.95	11.16	+73	72.00
	Thompson sampling	60	0.95	9.07	+40	72.00
	Thompson sampling + dropout model	60	0.95	6.74	+4	72.00
	UCB	60	0.95	9.28	+44	72.00
	Static α=0 (fixed depth)	60	0.95	6.45	+0	72.00
	Static α=1	60	0.95	5.84	−10	72.00
lowyield-c0	Round-robin + verified stop	60	1.00	7.58	–	15.47
	Coverage-greedy	60	1.00	8.29	+9	14.49
	Coverage-greedy + dropout model	60	1.00	7.34	−3	12.37
	Tail-aware	60	1.00	8.44	+11	18.72
	Thompson sampling	60	1.00	7.71	+2	13.45
	Thompson sampling + dropout model	60	1.00	7.42	−2	12.00
	UCB	60	1.00	8.39	+11	14.07
	Static α=0 (fixed depth)	60	1.00	7.58	+0	15.47
	Static α=1	60	1.00	8.62	+14	16.47
contam-c2	Round-robin + verified stop	60	1.00	3.90	–	5.71
	Coverage-greedy	60	1.00	4.47	+15	8.58
	Coverage-greedy + dropout model	60	1.00	4.00	+2	5.42
	Tail-aware	60	1.00	4.40	+13	6.30
	Thompson sampling	60	1.00	4.24	+9	6.56
	Thompson sampling + dropout model	60	1.00	3.97	+2	5.87
	UCB	60	1.00	4.47	+15	8.00
	Static α=0 (fixed depth)	60	1.00	3.90	+0	5.71
	Static α=1	60	1.00	3.77	−3	5.91
common-cause	Round-robin + verified stop	60	0.87	12.49	–	72.00
	Coverage-greedy	60	0.87	18.35	+47	72.00
	Coverage-greedy + dropout model	60	0.87	8.29	−34	72.00
	Tail-aware	60	0.87	20.08	+61	72.00
	Thompson sampling	60	0.87	17.08	+37	72.00
	Thompson sampling + dropout model	60	0.87	8.83	−29	72.00
	UCB	60	0.87	16.18	+29	72.00
	Static α=0 (fixed depth)	60	0.87	12.49	+0	72.00
	Static α=1	60	0.87	10.45	−16	72.00

at 48, with no wrong output at any setting. Part of the redundancy limit reported above is therefore a limit of *our decoder configuration*: a deeper consensus extracts more fragments from the surviving containers. Every zero- or partial-success result in this paper holds for this code, this placement, this consensus cap and this stream length, and a more patient decoder shifts the boundary.

Failure modes beyond total loss. Because total loss is the simplest failure to model, we added four further modes (Table XV): half of one container's designs physically lost before storage; a quarter lost in each of two containers; one container whose reads are 90% not derived from stored strands; another at 50%; and a common-cause scenario in which every container is aged two further half-lives and amplified harder. Three results stand out. First, partial physical loss is where the one-parameter model fails hardest: coverage-greedy costs +76.6% against round-robin, tail-aware +92.0%, UCB +59.7% — far worse than the +15% of the baseline. This is what the dropout explanation predicts, since a container missing half its designs is pure dropout. Second, the two-parameter

TABLE XVI

OTHER ARCHIVE GEOMETRIES, BASELINE CONTAINER CONDITIONS (20 SYNTHESES × 3 READ ORDERS; BIG: 10 × 3). M3/M12: 3 AND 12 CONTAINERS; N40/N64: $(32, 40)$ AND $(32, 64)$; K16: $(16, 24)$; BIG: A FOUR TIMES LARGER FILE. THOUSANDS OF READS.

Setting	Allocation	n	p_{ok}	$\bar{r}_{\text{ok}}$	vs. RR	q99
m3	Round-robin + verified stop	60	0.82	8.39	–	72.00
	Coverage-greedy	60	0.82	16.24	+94	72.00
	Coverage-greedy + dropout model	60	0.82	8.09	−4	72.00
	Tail-aware	60	0.82	20.27	+141	72.00
	Thompson sampling	60	0.82	14.56	+73	72.00
	Static α=0 (fixed depth)	60	0.82	8.39	+0	72.00
	Static α=1	60	0.82	7.50	−11	72.00
m12	Round-robin + verified stop	60	1.00	3.48	–	5.87
	Coverage-greedy	60	1.00	3.64	+5	6.68
	Coverage-greedy + dropout model	60	1.00	3.36	−3	4.85
	Tail-aware	60	1.00	3.63	+4	6.23
	Thompson sampling	60	1.00	3.67	+5	6.48
	Static α=0 (fixed depth)	60	1.00	3.48	+0	5.87
	Static α=1	60	1.00	3.27	−6	5.33
n40	Round-robin + verified stop	60	0.83	11.07	–	60.20
	Coverage-greedy	60	0.83	12.04	+9	60.20
	Coverage-greedy + dropout model	60	0.83	7.98	−28	60.20
	Tail-aware	60	0.83	13.55	+22	60.20
	Thompson sampling	60	0.83	11.90	+7	60.20
	Static α=0 (fixed depth)	60	0.83	11.07	+0	60.20
	Static α=1	60	0.83	10.59	−4	60.20
n64	Round-robin + verified stop	60	1.00	2.26	–	2.75
	Coverage-greedy	60	1.00	2.28	+1	3.08
	Coverage-greedy + dropout model	60	1.00	2.24	−1	2.82
	Tail-aware	60	1.00	2.32	+3	2.95
	Thompson sampling	60	1.00	2.29	+1	2.72
	Static α=0 (fixed depth)	60	1.00	2.26	+0	2.75
	Static α=1	60	1.00	2.17	−4	2.60
k16	Round-robin + verified stop	60	1.00	4.52	–	5.84
	Coverage-greedy	60	1.00	5.29	+17	8.64
	Coverage-greedy + dropout model	60	1.00	4.42	−2	6.12
	Tail-aware	60	1.00	5.11	+13	8.06
	Thompson sampling	60	1.00	4.82	+7	6.79
	Static α=0 (fixed depth)	60	1.00	4.52	+0	5.84
	Static α=1	60	1.00	4.22	−7	5.41
big	Round-robin + verified stop	30	1.00	15.26	–	20.05
	Coverage-greedy	30	1.00	17.43	+14	28.95
	Coverage-greedy + dropout model	30	1.00	15.62	+2	19.39
	Tail-aware	30	1.00	17.16	+12	27.61
	Thompson sampling	30	1.00	16.59	+9	23.34
	Static α=0 (fixed depth)	30	1.00	15.26	+0	20.05
	Static α=1	30	1.00	14.49	−5	18.42

model does more than remove the penalty: coverage-greedy with dropout reaches -6.0% under partial loss and -33.6% under common-cause degradation, the largest allocation savings anywhere in our digital-twin experiments. Third, success probability again does not depend on the policy (0.95 with two quarter-lost containers, 0.87 under common-cause degradation, identical across rules), and no configuration produced a wrong output. Low yield and contamination are mild by comparison ($+9.3\%$ and $+14.6\%$ under the one-parameter model, -3.2% and $+2.4\%$ with dropout).

Other archive geometries. Our primary design is one geometry, so we repeated the baseline conditions on six others (Table XVI): 3 and 12 containers, $(32, 40)$, $(32, 64)$ and $(16, 24)$ codes, and a file four times larger. Every qualitative conclusion survives all six. Coverage-greedy under the frozen model is never cheaper than round-robin (from $+1.1\%$ at $(32, 64)$ to $+93.5\%$ with three containers); with the dropout model it is never materially worse (-27.9% to $+2.4\%$); the hindsight static plan with stopping is cheaper than round-robin everywhere (-4.0% to -10.7%); and there was no wrong output in any geometry. The magnitudes vary strongly and sensibly: with high

TABLE XVII

READS ARE NOT THE ONLY COST. TWO FURTHER CHARGING MODELS APPLIED TO THE SAME BASELINE RETRIEVALS; THE READS-ONLY MODEL IS THE READS COLUMN ITSELF. A MULTIPLEXED RUN CHARGES 50 READ-EQUIVALENTS PER BATCH OF 50 READS AND 500 PER CONTAINER LIBRARY PREPARATION; AND A SEPARATE SEQUENCING RUN PER BATCH (2,000 PER BATCH). COSTS IN READ-EQUIVALENTS (THOUSANDS). SWITCHING CONTAINERS EVERY BATCH IS FREE IN THE FIRST MODEL AND DOMINANT IN THE THIRD.

Allocation	Reads	Batch.	Sw.	multiplexed	run/batch
Static α=1	3.31	66	65	9.62	138.72
UCB + dropout model	3.37	67	57	9.74	141.09
Static α=0 (fixed depth)	3.48	70	69	9.96	145.69
Round-robin + verified stop	3.48	70	63	9.96	145.69
Thompson sampling + dropout model	3.52	70	54	10.03	147.15
Coverage-greedy + dropout model	3.55	71	57	10.09	148.40
Thompson sampling	3.72	74	56	10.45	155.66
Pivotal + dropout model	3.75	75	56	10.49	156.63
Tail-aware + dropout model	3.82	76	38	10.64	159.72
Tail-aware	3.92	78	38	10.85	163.88
Learn-then-commit + dropout model	3.93	79	59	10.87	164.29
Coverage-greedy	3.95	79	61	10.89	164.78
UCB	4.00	80	54	11.00	166.96
Info-greedy + dropout model	24.84	497	44	52.68	1021.42

redundancy $(32, 64)$ the whole allocation question nearly vanishes ($+1.1\%$, and only 3.4k reads of stream are ever needed), while with three containers — a third of the archive near-dead instead of a sixth — it is largest. Recoverability also varies: success is 1.00 in four geometries but 0.82 with three containers and 0.83 at $(32, 40)$, uniformly across policies, which is the redundancy effect and not a policy effect.

M. Reads are not the only cost

Our frozen metric is reads, which is the right unit for a pay-per-read service and the wrong one for a laboratory that pays per run. Table XVII reprices the same baseline retrievals under three charging models: reads only; a multiplexed run that charges 50 read-equivalents per batch of 50 reads and 500 per container library preparation; and a separate sequencing run per batch at 2,000 read-equivalents. Two things follow. The ranking of the policies is unchanged — round-robin and the tuned static plans stay cheapest, information-greedy stays catastrophic — because every policy's batch count is proportional to its read count, so a per-batch charge scales all of them together. But the *margin* that allocation is competing for shrinks relative to the fixed charges: under the per-batch-run model the whole spread between round-robin and the best adaptive rule is a few percent of a bill dominated by run setup. We also record container switches, which the frozen runs did not. At a batch size of 50 the adaptive rules switch container 38–61 times per retrieval (tail-aware fewest at 38, coverage-greedy most at 61), round-robin 63 times, and the fixed-proportion plans most of all (65 for α=1, 69 for uniform), because they cycle to hold their proportions. No policy here is switch-frugal by design, and a setting where switching is expensive would need a larger batch or an explicit switch penalty in the score; the rule that happens to switch least, tail-aware, is also among the costliest in reads, so no policy here deliberately trades switches for reads.

Three quantities are distinct: reads *consumed* (what a policy spends), capacity *reserved* (what a non-stopping plan must commit in advance), and laboratory *expenditure* (what an invoice says). Our 27–32% figure compares the first with the second; it is *not* a demonstrated 27–32% reduction in the cost of sequencing. Converting it would require per-read pricing with no fixed component, which no laboratory we know of offers. We did not model instrument

time, reagent costs, library preparation, minimum run sizes, flow-cell granularity, or the cost of the calibration reads used by the replay's calibrated belief (charged separately in Section VI-K). Under the charging models we did examine, the fixed components dominate. The read counts are the measured quantity; the monetary columns only illustrate how little of a read saving may remain once fixed charges apply.

N. Supporting channel result (real nanopore reads)

On two independent nanopore flowcells of DNAformer reads [42], [51], a strand-specific error propensity persists across flowcells (r=0.57; null with matched contexts 0.002), and a hierarchical beta-binomial error model fitted on one flowcell predicts consensus success on held-out strands of the other with mean absolute error 0.083, against 0.181 for an i.i.d. model. This motivates over-dispersed models but does not enter the controller.

VII. Discussion

What helps, and whom. The robust gain in our experiments is verified stopping: reading until every stripe, its CRC-32 and the file hash verify, and then stopping. It needs no model of the containers, never returned a wrong file, and for a user who pays per read it turns the capacity a static plan must reserve into a bill 27–32% smaller. It does *not* reduce the capacity a user must reserve, because the q99 is unchanged. Choosing *which* container to read next is a second-order effect in every configuration we tested: worth a few per cent at best, and strongly negative when the container model cannot express what is wrong with a container. The two best cases reported in this paper come from *different experiments* and should not be added. The 4.2% saving is the pre-specified *fixed* allocation, which runs no controller and whose proportions can be estimated offline, measured against round-robin with both stopping on the baseline. The 3.5% saving is UCB, an *online* rule, measured against the same round-robin on the same baseline but only after the container model is extended with a dropout parameter, a secondary configuration that is not the frozen one. They are two routes to a similar small gain, not a cumulative 7.7%. In practice, the decision most worth engineering is *when to stop*; the next is the *container model*, not the allocation rule built on it.

Calibration versus cost, and model versus likelihood. The likelihood that gave the best-calibrated intervals (ρ_U=0.1) also gave the costlier allocations, and the binomial likelihood, whose intervals covered only 54%, gave cheaper ones. Calibrating the uncertainty of a misspecified model does not improve its decisions. The later experiments separate the two questions. On *calibration*, re-selecting ρ_U with the prior actually used and on a wider grid still cannot cover the near-dead container at any value; that failure belongs to the container model, and only a dropout parameter addresses it. On *cost*, the likelihood matters a great deal: at the re-selected ρ_U=0.03 most of the allocation penalty disappears without changing the model (Section VI-J). A configuration can therefore be poorly calibrated and still allocate acceptably, and our strongest negative result came from a setting that was poor on both counts. The practical priority is to repair the container model before tuning the posterior over it. We therefore describe the controller as online-adaptive rather than self-calibrating, and do not present its forecasts as probabilities.

Digital twin versus replay. The two settings agree that stopping is the main saving and that no adaptive rule lowers q99 reliably. On allocation with the generic prior they agree where the conditions match. On the Twist pool, whose containers are runs of one synthesis batch with no near-dead member (the mild-heterogeneity end of the sweep), allocation saved 2–3%; on the Genscript pool, which has a near-dead container, it cost 0–3% and over-read that container, as on the digital twin. Only prior knowledge of the weak container, obtained from calibration reads, turned allocation into a large saving.

Fig. 6 summarises the practical recommendations that our results support for archives like ours.

Future experimental validation. Physically separate containers would allow three tests. (i) Whether real containers show the near-dead, dropout-dominated behaviour that made adaptive allocation lose here: one design would be synthesised in several independent batches, aged and amplified differently, and each container sequenced to saturation to measure its fraction of absent strands. (ii) Whether verified stopping keeps its saving and its safety on real sequencing runs with our own inner code: reads from a nanopore or a split Illumina run would be streamed through the same decoder, counting wrong outputs and inner-code false accepts against the known file. (iii) What real container failures look like and how often they occur: a set of containers would be tracked over accelerated

Practical recommendation (multi-container retrieval).

1) **Read the containers in turn and stop on verification.** Check every stripe against its CRC-32 and the file against its hash. This requires no container model and saved 27–32% relative to a reserved budget. An unverified file should never be returned.
2) **Online adaptation does not help users who reserve capacity.** A user who reserves capacity rather than paying per read should budget for the q99 of round-robin. Fixed proportions tuned to the containers did lower the tail (by about 9% here), so effort is better spent on measuring the containers once than on an adaptive controller.
3) **Deprioritise, but do not permanently exclude, a container that appears dead.** In our simulated total-loss scenario, where a container returns no fragment derived from stored strands, posterior rules reduced the cost of the retrievals that still succeeded by up to 24%, but did not make recovery more likely. A probe that yields no verified fragment does not show that a container is physically dead. Low efficiency, partial strand loss or a need for deeper consensus can look the same after one batch, yet behave very differently (Sections VI-F and VI-L): with partial loss the file remained recoverable from that container, and raising the consensus cap alone moved the one-failed-container success rate from 0.313 to 0.357. Because we did not design or validate a failure-detection rule and report no false-exclusion rate, we recommend down-weighting such a container while continuing to sample it.
4) **Do not use a model-based allocation rule with a container model that cannot represent dropout.** Every such rule we tested read 7–441% more than round-robin. With efficiency *and* dropout (a different, non-frozen configuration), the best online rule, UCB, gained 3.5% on the baseline, which is comparable to the fixed plan and not additional to it.
5) **Calibrate only for archives that will be read repeatedly.** A prior fitted on held-out reads of the same containers identified the weak container in advance and saved about 16% per retrieval on the Genscript replay, the largest allocation effect we measured. It consumed 1.8M calibration reads, 2.8 times the cost of one retrieval, and broke even only after roughly 17 retrievals (about 10 at 100k calibration reads per container). Smaller calibration budgets did not help consistently, and 30k reads per container was *worse* than no calibration. This result comes from a semi-empirical replay with oracle decodability. Estimating only the static read proportions is a cheaper route to most of the available gain.
6) **Distinguish read savings from laboratory-cost savings.** Every number above is a read count, and a 27% or 16% reduction in reads does not imply the same reduction in sequencing cost. Instrument time, reagents, library preparation, minimum run sizes and real switching costs are not modelled, and under our assumed per-batch and per-container charges the margin available to allocation shrinks sharply (Section VI-M).
7) **Size redundancy for the failures the archive must survive.** With $(32, 48)$ over six containers, two lost containers leave exactly K fragments per stripe and no margin, and under a noisier synthesis model the file was unrecoverable in our design regardless of policy.

Fig. 6. Practical recommendation derived from Section VI.

aging, which would calibrate the failure prior and the redundancy needed per container. The decisive experiment, not performed here, is the full chain on physical material: encode a known file with these codes, synthesise it, split it into independently synthesised and separately handled containers, sequence each one, feed the FASTQ reads into this decoder, and compare the returned bytes with the original. Until then, the replay in Section VI-K remains semi-empirical and the end-to-end evidence remains simulated.

VIII. Limitations and Threats to Validity

Table XVIII lists the main threats. *External validity*: every result is computational. The digital twin uses DT4DDS's default parameters and one archive geometry (1,200 strands, $(32, 48)$, six containers), and its failure mode is simulated (a failed container returns non-informative reads). The replay uses real reads, but decodability is an oracle event and the "containers" are runs of one synthesis batch. *Internal validity*: ρ_U was selected with prior sd $s{=}1$ and used with $s{=}0.5$; the selected value lies at the edge of the pre-specified grid; and one code fix (static allocator) was made after freezing (Appendix B). The photolithographic scenario was first run at a reduced design and re-run at the full 30×10 design for round-robin, with the same outcome. The pre-specified static baselines use efficiencies measured in hindsight on the evaluated syntheses, which favours them; the calibration-estimated

variant, which sees no evaluation data, performs the same within noise, so that advantage is not what drives the static result. *Construct validity*: read counts are a proxy for cost; we do not model run set-up costs, which favour fewer, larger batches. *Statistical*: wrong-output and false-accept bounds come from zero events and scale with the number of trials; the comparisons are paired but within one archive design. We prove no optimality or regret result.

IX. Reproducibility and Availability

Artifact. Code, configuration, every result file and the scripts that build every table and figure are in the public repository https://github.com/SaiMouli3/Dev (directory `d3ra-dna`), at the commit recorded in `research_paper/final/COMMIT.txt` of that revision. The environment is Python 3.13.18 with NumPy 2.5.3, SciPy 1.18.1, pandas 3.0.5 and DT4DDS 1.1.0 (pinned in `requirements.txt`); the public data are ENA PRJEB65931 [46] (Illumina) and the DNAformer nanopore reads [51], [52], both fetched by the scripts in `experiments/`. `research_paper/audit/` holds the claim-to-number recomputation and its output.

What reproduces what. The digital-twin experiments are driven by `experiments/dt_experiments.py` (sub-commands `gen`, `run`, `runx`, `static`, `calib`, `calib_eval`, `forecast`, `posthoc_rho0`), the replay by `experiments/paper_experiments.py` (`replay_v3`, `replay_calcost`); statistics by `dt_analysis.py` and `r4_analysis.py`; tables and figures by `make_tables.py`, `make_r4_tables.py` and `make_dt_figures.py`, which read only committed result files, so no number in any table or figure is typed in by hand. Table XIX maps each claim to its command and result file, and `dt_protocol.yaml` and `dt_protocol_rev4.yaml` record the frozen protocols and every deviation.

Determinism. Our own layer is deterministic: every random draw in placement, read ordering, the controller and the decoder is seeded by CRC-32 of a string naming the scenario, replicate, container, read order and policy, so those results do not depend on worker count or scheduling, and 24 sampled baseline retrievals re-executed single-process in reverse order reproduce every recorded field (status, reads, per-container reads, rounds, decode attempts, false accepts, verification rounds) exactly. The *simulator* is not: DT4DDS 1.1.0 does not produce identical read streams from a fixed seed, and calling its synthesis–aging–PCR–sequencing chain twice in one process with the same seed gives different streams. The committed read streams are therefore part of the artifact, and a reader who regenerates them does not get our retrievals back.

We quantified the consequence. Regenerating all 30 baseline replicates from a clean clone of the final commit and re-running all eight policies and four static plans (3,600 retrievals) leaves every outcome identical — the same successes, the same zero wrong outputs — while 2,350 of 2,400 retrievals consume a different number of reads (mean absolute difference 1,629). The conclusions are unchanged: every policy's mean falls inside the other run's cluster-bootstrap interval in both directions, the ordering of all twelve policies is identical, and the headline comparisons move by one to three percentage points (round-robin 3.49k against 3.51k; the hindsight static plan -4.2% against -5.4%; coverage-greedy $+15.1\%$ against $+12.2\%$). The full comparison is in `evidence/verification_reports/clean_repro.txt`, regenerated by `clean_repro.sh` and `clean_repro_compare.py`. At condition level, the Genscript replay conditions re-run in separate parallel processes reproduced the sequential run's values exactly, because that pipeline consumes fixed real reads rather than simulated ones.

The severity sweep, the failure modes, the six alternative geometries, the photolithographic scenario and the two replays were *not* re-executed for the final build; they are reproduced from their committed result files and seeds. A full re-execution takes several CPU-days, dominated by DT4DDS synthesis and by the information-greedy policy, and the commands are in the artifact.

X. Conclusion

On a digital twin whose containers are independently synthesised *in simulation*, verified stopping returned the stored file without a single wrong output in 62,310 retrievals and reduced the reads actually consumed to 27–32% below the capacity a non-stopping plan must reserve. This saving requires no model of the containers and is the main result of the paper.

Deciding *which* container to read next is worth far less. Fixed proportions, which run no controller and can be estimated offline from calibration replicates, gave 4.2% on the baseline. Under the frozen container model

TABLE XVIII

ASSUMPTIONS, LIMITATIONS AND THEIR CONSEQUENCE. ITEMS MARKED † ARE TESTED EXPERIMENTALLY IN THIS PAPER; THE RELEVANT RESULT IS CITED.

Limitation	Consequence
Computational only (digital twin + replay)	no wet-lab confirmation; DT4DDS fidelity bounds external validity; future experiments are outlined in Section VII
Single efficiency multiplier θ_j in the frozen model	cannot represent dropout; adaptive allocators over-read near-dead containers. † A two-parameter model (efficiency and dropout) removes most of the penalty (Section VI-F)
ρ_U at the edge of its grid; selected with s=1 while runs use s=0.5	† re-selected with s=0.5 on an extended grid: no value reaches 0.90 coverage and the near-dead container is never covered (Table X); † the complete primary comparison re-run at the re-selected ρ_U=0.03 weakens the allocation penalty from harmful to merely no better for four of seven rules (Table XI)
Calibration-optimal vs. cost-optimal ρ_U differ	a better-calibrated posterior allocated worse; the ablation separates likelihood, prior and model (Table VIII)
Gains are in the mean, not the tail	† no *adaptive* rule lowered q99 (a hindsight static plan did, by about 9%); tail intervals are wide and reported (Table IX)
Failures simulated	real failure modes and rates unknown; † partial loss, low yield, contamination and common-cause degradation are now tested (Table XV), but their rates are still assumed
One archive geometry in the primary design	† six further geometries tested (Table XVI); conclusions outside those remain untested
Consensus cap of 24 reads per cluster	† caps of 12 and 48 tested (Table XIV); decoder configuration does affect the recovery boundaries
Replay is semi-empirical	oracle decodability, so it is *not* end-to-end exact retrieval; containers are runs of one synthesis batch, not independent syntheses
Genscript pool: 472 of 12,472 designs excluded	4.23% of mapped evaluation reads discarded; identical material in every condition
Read count is not money	† per-batch and per-container charges change the ranking (Table XVII); instrument time, library preparation and minimum run sizes are not modelled
Container choice every 50 reads	an allocation policy over independent read streams, not a demonstrated physical sequencing controller (Section IV-C)
Safety bounds assume independence per retrieval	† retrievals share syntheses; the conservative per-synthesis bound is two orders of magnitude weaker (Table IV)
Nanopore correlation evidence: two flowcells	preliminary supporting evidence only; does not validate the container model
Information-greedy on the Genscript replay did not complete	reported as censored, not as a result (Appendix B)
DT4DDS 1.1.0 is not seed-deterministic	regenerated read streams differ, so the committed streams are part of the artifact; † a clean re-simulation reproduces every mean, paired comparison and the policy ordering, not the individual retrievals (Section IX)
No theory (regret, optimality)	empirical claims only, about the configurations tested

TABLE XIX

CLAIM-TO-EVIDENCE SUMMARY (FULL TABLE: `evidence/evidence_traceability.csv`; EVERY NUMBER RECOMPUTED FROM THESE FILES BY `audit/recompute_claims.py` AND `r4_analysis.py`). DT: DT4DDS DIGITAL TWIN, FULL DECODE CHAIN, NO ORACLE. REPLAY: REAL READ STREAMS, ORACLE DECODABILITY. NO RESULT IN THIS PAPER IS WET-LAB EVIDENCE.

Claim	Command (`dt_experiments.py` unless noted)	Result file (`results/`)	Evidence
No wrong output, no false accept; cluster-aware bounds (Table IV)	all DT runs	`dt_*.csv` → `r4_safety.csv`	DT
Stopping converts a 4.80–5.15k reservation into a 3.49k bill (Table V)	`run`/`static baseline`	`dt_baseline{,_static}.csv` → `r4_actual_vs_reserved.csv`	DT
Fixed proportions with stopping beat round-robin: -4.2% (pre-specified α=1), -4.6% (extended grid optimum α=1.5), -4.4% (calibration-estimated) — all mutually indistinguishable	`static baseline` (+`_r4fine`, `_r4cal`)	`dt_baseline_static{,_r4fine,_r4cal}.csv`	DT
Model-based rules costlier than round-robin under the frozen configuration (7–441%)	`run baseline`	`dt_summary.csv`	DT, negative
Penalty depends on the likelihood: at the re-selected ρ_U=0.03, four of seven rules are indistinguishable from round-robin, three still worse (Table XI)	`runx SCEN _rho003`	`dt_*_rho003.csv` → `r6_likelihood_sweep.csv`	DT, qualifies the above
The penalty needs the frozen *combination* of container model, likelihood and prior width; relaxing any one of the three substantially reduces it. Adding dropout is the physically meaningful change and helps most where loss dominates (Table VIII)	`runx SCEN _r4drop`	`dt_*_r4drop.csv` → `r4_ablation.csv`	DT
Likelihood, prior and model separated by ablation	`runx baseline _r4abl_*`	`dt_*_r4abl_*.csv`	DT
After an outright failure: -24% on successes, -3% per attempt, success unchanged (Table VII)	`run fail1`	`dt_fail1{,_static}.csv`	DT, simulated failure
Crossover with severity of the worst container	`run sweep_cov*`	`dt_sweep_cov*.csv`	DT
No *adaptive* rule lowers q99 *reliably* (three dropout-model configurations are 1–9% below numerically, intervals contain zero; fixed proportions -8.7% with the interval excluding zero) (Table IX)	all DT runs, `_r4ext`	`r4_tails.csv`	DT, negative
ρ_U re-selected at s=0.5 on an extended grid; near-dead never covered (Table X)	`calib`, `calib_eval` (s=0.5)	`dt_calibration_*_r4sd05.csv`	DT, negative
Forecasts over-confident, worse than climatology	`forecast baseline`	`dt_forecast_baseline.csv`	DT, negative
Prior sensitivity (failure rate, dropout, shape, grid, s, φ)	`runx baseline _r4prior_*`	`r4_prior_sensitivity.csv`	DT
Failure modes: partial loss, low yield, contamination, common cause (Table XV)	`runx {partial,lowyield,contam,common}*`	`r4_failure_modes.csv`	DT, simulated
Six further archive geometries (Table XVI)	`D3RA_GEOM=... runx baseline _r4`	`r4_geometries.csv`	DT
Consensus cap 12/24/48 (Table XIV)	`runx ... _r4cap{12,48}`	`r4_consensus_cap.csv`	DT
Homogeneous control; photolithographic boundary	`run homog`, `run photolitho`	`dt_{homog,photolitho}{,_static}.csv`	DT
Cost model with per-batch and per-container charges (Table XVII)	from committed per-retrieval rows	`r4_cost_model.csv`	DT, assumed prices
Leak-free real-read replay, both pools	`paper_experiments.py replay_v3`	`replay_v3{,_genscript}.csv`	Replay
Calibrated prior: -16% per retrieval, but 1.8M calibration reads, so ≈17 retrievals to break even (≈10 at 100k reads per container; lower levels save nothing) (Table XIII)	`paper_experiments.py replay_calcost`	`r4_replay_calcost.csv`	Replay
Leaky first analysis (upper bound only)	`real_data.py policies`	`realdata_*policies*_v2.csv`	superseded
Correlated nanopore errors (MAE 0.083 vs. 0.181)	`q3_correlated_errors.py`	`q3_summary.json`	real nanopore reads
Regret bounds; wet-lab containers; real failure rates; independently synthesised physical containers	–	–	not attempted

and likelihood, every online rule cost *more* than round-robin when a container was near-dead, by 7–441%. This penalty is a property of the pre-registered configuration: at the likelihood chosen by a corrected selection, four of the seven rules become indistinguishable from round-robin rather than harmful. With a dropout-aware model, another secondary configuration, the best online rule reached a comparable 3.5%; the two best cases come from different experiments and do not add up. No adaptive rule beat round-robin by more than a few per cent in any combination we tested. The ablation shows that the penalty requires the frozen combination of container model, likelihood and prior width, and that relaxing any one of the three removes most of it. A dropout parameter is the physically meaningful repair and is most useful where loss dominates, reaching -6% under partial loss, -34% under common-cause degradation and -35% after an outright failure. No adaptive rule reduced the 0.99-quantile

TABLE XX

UPPER BOUND WITH EVALUATION LEAKAGE (SUPERSEDED FIRST ANALYSIS, KEPT FOR TRANSPARENCY). REPLAY OF REAL READ STREAMS WITH "CALIBRATED" BELIEFS WHOSE CONSENSUS CURVES WERE POOLED FROM *evaluation*-SPLIT READS (APPENDIX A); THOUSANDS OF READS AT 0.99. THESE NUMBERS ARE NOT USED FOR ANY CLAIM IN THE MAIN TEXT. STATIC PLANS PAY THEIR BUDGET ON EVERY RETRIEVAL; FOR ADAPTIVE POLICIES THE 0.99-QUANTILE Q99 IS SHOWN SEPARATELY. Δ: ADAPTIVE (PIVOTAL) MINUS ORACLE-TUNED STATIC, PAIRED (95% CI).

Pool	Placement	Fixed depth	Oracle-tuned static	Adaptive stop, round-robin	Adaptive, cov.-greedy	Adaptive, pivotal	q99 of pivotal	Δ (95% CI)
Twist	round-robin (6)	67.9	63.1	57.8	55.7	55.8	63.2	-7.3 (-12.1, -7.0)
Twist	reliability-weighted	65.4	61.9	56.0	54.2	54.1	61.8	-7.9 (-13.3, -7.6)
Twist	tiered (4)	61.9	53.1	51.3	46.4	46.2	53.0	-6.9 (-19.1, -6.2)
Twist	single best	53.9	53.9	44.8	44.8	44.8	54.0	-9.1 (-11.0, -6.7)
Genscript	round-robin (6)	810.4	617.9	639.9	532.1	531.9	694.2	-86.0 (-127.1, -78.8)
Genscript	single best†	279.4	279.4	219.0	219.0	219.0	279.6	-60.4 (-70.5, -54.2)

200 read permutations, except †100. All policies completed in every trial. The single-best container is chosen with the calibrated belief, which contains the evaluation-split leak of Appendix A. Superseded first-run Genscript numbers (computed before the per-strand consensus cache was regenerated) are kept in `results/realdata_genscript_policies_{roundrobin,part2}.csv`.

in any configuration we tested, so none of these gains lowers reserved capacity. The posterior is not calibrated for a near-dead container at any value in its likelihood family, and the completion forecasts are worse than climatology and are not usable probabilities.

The digital-twin experiments run the full decode chain and compare bytes; the real-read replay uses public Illumina streams but treats decodability as an oracle event and emulates containers by subsetting runs of one synthesis batch. No DNA was synthesised or sequenced for this paper, and each headline figure is a read count, not a laboratory cost. The paper provides a regenerable baseline, bounds on wrong outputs that account for their dependence structure, and an account of when online adaptation hurts. The open questions require physically independent containers with measured failure and dropout statistics, and an end-to-end retrieval of a known file from synthesised DNA through this decoder.

APPENDIX A
UPPER BOUND WITH EVALUATION LEAKAGE

Our first analysis of the real read streams (Table XX) used a "calibrated" belief whose per-strand consensus curves were pooled over reads that included the evaluation split. The controller therefore had information about the very reads it was evaluated on, and the static plans were tuned on them too. We keep the table only as an *upper bound with evaluation leakage*; no claim in the main text uses it. In the leak-free re-analysis (Table XII) the primary belief is the generic prior, and the optional calibrated belief uses only the first 300k reads of each run while evaluation uses the next 900k. On Twist, the leaky analysis put coverage-greedy at 55.7k reads and the oracle-tuned static plan at 63.1k; the leak-free analysis gives 56.7k and 65.7k. The leak therefore flattered the adaptive policy by about 1k reads and the static baseline by about 2.6k, and the leak-free gap is larger, not smaller.

APPENDIX B
PROTOCOL DEVIATIONS AND POST-HOC ANALYSES

Deviations. The frozen protocol and every change made after freezing are in `dt_protocol.yaml`. (i) The planned CustomArray scenario was replaced, before any policy run, by DT4DDS's photolithographic synthesis model, because the CustomArray settings carry a 117-nt position-bias table that cannot be applied to 136-nt strands. (ii) We suspected from commit timestamps that the first Twist replay had run with $\rho_U{=}0$ and re-ran it; a direct test showed that it had used the frozen $\rho_U{=}0.1$, and the re-run reproduced it exactly, so nothing changed. (iii) One cached homogeneous replicate was truncated by an interrupted job; it was regenerated from its seed and the scenario re-run in full. (iv) The photolithographic scenario was first run with 10 syntheses $\times$ 3 read orders for compute reasons; the first 50 retrievals of the full run were discarded unseen. Revision 4 re-ran round-robin on it at the full 30 $\times$ 10 design, with the same outcome (0 of 300 recovered). (v) The static allocator crashed when

only zero-weight (failed) containers remained; it now reads them out, which can only affect failed retrievals of the failure scenarios. The ρ_U rule was applied as frozen even though the selected value lies at the edge of the grid. (vi) The sequential Genscript replay finished information-greedy after about six hours but then crashed while computing its paired difference, because some trials never completed (infinite cost); nothing was saved. We fixed the bug, ran the other Genscript conditions in parallel processes with the same per-condition seeds (they reproduced every value logged by the sequential run), and re-started information-greedy; it had not finished when this version was built, so Table XII has no information-greedy row for Genscript. What is known about that condition is that in the sequential run its 200 trials finished and more than 1% of them never recovered the file within the available reads, which is why the 0.99 budget was infinite and the paired statistic overflowed. We treat the condition as *censored*: it is excluded from every average and comparison and marked as not completed in Table XII. Not every policy therefore completed every real-read condition. (vii) Revision 4 adds the pre-registered secondary analyses of `dt_protocol_rev4.yaml`: the dropout-aware container model, the $2 \times 2 \times 2$ ablation, the ρ_U re-selection at $s{=}0.5$ on an extended grid, prior sensitivity, five further failure modes, six further archive geometries, consensus caps of 12 and 48, finer and calibration-estimated static families, an extended baseline replicate set, the full-design photolithographic run, the replay calibration-cost experiment and the cost model. The frozen revision-3 settings remain the primary configuration; every revision-4 result is secondary and is reported whatever its direction.

Post-hoc analyses (not pre-specified). With the binomial likelihood ($\rho_U{=}0$), on the same 300 baseline retrievals, coverage-greedy used 3.34k reads (-4.4% vs. round-robin, against $+15.1\%$ at $\rho_U{=}0.1$), Thompson sampling 3.37k (-3.5%, against $+7.4\%$), tail-aware 3.37k (-3.5%, against $+12.9\%$) and pivotal 3.64k ($+4.4\%$, against $+49.3\%$); q99 was 4.95–5.70k, still not below round-robin's 5.15k. With one failed container the binomial likelihood helped tail-aware (-25% vs. $+2\%$) and Thompson sampling (-20% vs. -9%) and hurt coverage-greedy and pivotal slightly (-15% vs. -17%; -8% vs. -15%). Success and safety were unchanged (no wrong output). On the Twist replay, the binomial likelihood made coverage-greedy and Thompson sampling costlier (57.6k each, against 56.7k and 56.9k at $\rho_U{=}0.1$), the opposite of the digital-twin result: the two settings disagree on which likelihood allocates better.

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